Welcome to

COMPUTER NETWORKS

The Invisible Technology Behind Everything You Build

> Behind every AI model, cloud application, online payment, and cyberattack lies a computer network.

How many networks have you already used before entering this classroom?



CONNECTING PEOPLE | POWERING APPLICATIONS | ENABLING INNOVATION | SECURING THE FUTURE

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How Many Networks you are using Today!

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Course Title: Computer Networks



Course Code: CSE3006

Course Details

Nature of the course: LTP

L- Lecture ; T- Tutorial ; P- Practical

Credits: 4



Syllabus: Unit 1

Introduction: Networking goals, Networking Topologies, Network Architecture - Interfaces and services, ISO-OSI Reference Model, TCP/IP Protocol suite. Comparison of OSI and TCP/IP.

Signals: Analog Signals and Digital Signals, Characteristics of Analog Signals, Frequency Modulation (FM), Amplitude, Modulation (AM), Phase Modulation (PM), Quadrature Amplitude Modulation QAM.

Encoding and Modulation: Digital to Digital signal conversion, Digital to Analog signal conversion, Analog to Analog signal conversion, Analog to Digital signal conversion

Unit 2

Physical Layer: Transmission Media - Guided Transmission Media – Twisted pair, Coaxial Cable, Fiber Optics , Unguided Media–Radio waves, Microwaves, Infrared.

Switching: Packet and Circuit Switching

Data Link Layer: Framing, Flow and Error Control, Error Detection and Correction - Types of Errors, Redundancy, LRC, CRC, Checksum. Flow and Error control Protocols: Stop and Wait, Stop and Wait ARQ, Go back-N ARQ, Selective repeat ARQ.

Unit 3

Local Area Network: Ethernet: Multiple Access Links and Protocols –ALOHA, CSMA, CSMA/CD, CSMA/CA. 802.3 IEEE standard, Token Bus: 802.4 IEEE standard, Token Ring: 802.5 IEEE standard, Wireless LAN Protocols.

Logical Addressing: IPv4 Addresses, Ipv6 Addresses. INTER Networking Devices: Layer 1 connections- Repeater, Hubs, Layer 2 connections- Bridges, Switches, Layer 3 connections- Routers, Gateways.

Unicast Routing Protocols: Distance Vector Routing, Link State Routing. Internet Protocols: Class A, B, C, D, E traffic, IPV4, IPV6. Address Mapping: ARP, RARP- ICMP- IGMP.

Unit 4

Transport Layer: Services, Multiplexing and Demultiplexing, Process to process delivery –User Datagram Protocol (UDP) – Transmission Control Protocol (TCP) – Congestion Control Techniques– Congestion Prevention Policies - Quality of services (QoS) – Techniques to improve QoS -Integrated Services

Unit 5

Principles of Networking Applications: Web, FTP, E-mail, DNS, Peer-to-Peer applications, Creating Networking Applications, Telnet, SNMP, Recent trends in Computer Networks.

Recent Trends in Network Security: Cryptography, Substitution and Transposition Ciphers, Data encryption standard (DES), breaking DES, Public Key Cryptography, Network layer security (IPSec), Transport Layer Security (TLS/SSL, HTTPS)

Course Outcome

The Students who will be able

- To understand the basic principles of Communication, Networks and Satellite Transponders.
- To understand bit rate and baud rate.
- To understand the concepts of CSMA/CD, ISDN.

Contd...

- To design sub-netting and enhance the performance of routing mechanisms.
- To compare various congestion control mechanisms and identify suitable Transport layer protocol for real time applications.
- To identify various Application layer protocols for specific applications

Textbooks

1. Larry L.Peterson, Bruce S.Davie, Computer Networks: A System Approach, Fifth Edition, 2012.
2. James F.Kurose and Keith W.Ross, Computer Networking: A Top-Down Approach, Sixth Edition, Pearson Education, 2013.

References

1. Behrouz A. Forouzan, Data Communications and Networking, Fifth Edition, McGraw-Hill, 2012.
2. Behrouz A.Forouzan, TCP/IP Protocol Suite, 5th Edition, Tata McGraw-Hill, 2012
3. W.Richard Stevens, TCP/IP Illustrated, The Protocols, 2nd Edition, Prentice Hall 2012
4. Andrew S.Tenanbaum, Computer Networks, Fifth Edition, Prentice Hall, 2012

Continuous Assessment

Assessment Components	Number	Weightage
Class Assessment - Tutorial	3 tutorials	5%
Class Assessment – Group Activity	1 activity	5%
Lab Assessment – Continuous Assessment	1 per each experiment	10%
Lab Assessment – Challenging Task	One task	10%
Lab Assessment – Viva	2 Quiz	5%

Continuous Assessment

Assessment Components	Mode	Weightage
Continuous Assessment Test 1 (2 Modules)	Closed Book	15%
Continuous Assessment Test 2 (2 Modules)	Open Book	15%
Term End Examination (Full Portion)	Closed Book	30%

Ground rules for the course

- The teaching learning process in the course will be interactive and transparent.
- The students can feel free to ask **any clarifications** regarding the course can **post the question in Google Classroom**/ email.
- Students are expected to be **regular and punctual to the classes-** late comers (more than 5 minutes) can attend the class but will be marked absent for the hour.
- Submission of Tutorial and other task related to course should be completed before the deadline.

Ground rules for the course

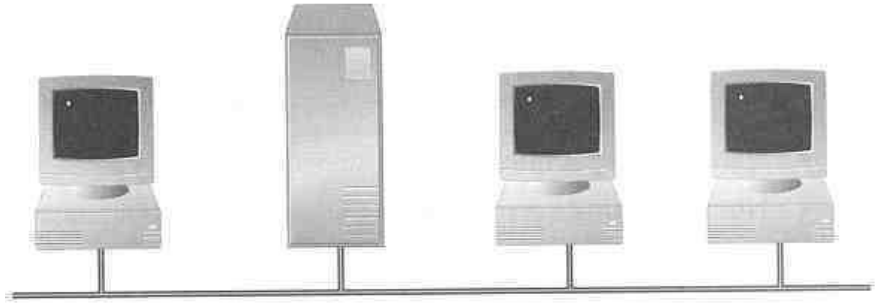
Plagiarism Policy

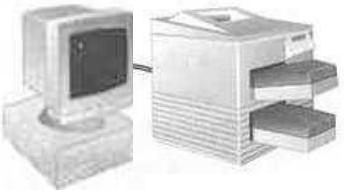
- Students are expected to avoid plagiarism in assessments that are planned for the course.
- Plagiarism if found any, the students involved (student copied and student helped to copy) has to resubmit for first time.
- If found repeated for the second time, his / her documents will not be evaluated and zero marks for submission will be awarded.

Introduction

Introduction

- A network is a set of devices (often referred to as nodes) connected by communication links.
- A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.





Computer Network

- A **computer Network** is defined as a set of computers interconnected by a communication links for sharing data or resources or exchanging messages.
- Communicate – Sharing Information
- Sharing can be Local or Remote
  - Local Communication – Happens Face to face
  - Remote Communication – Takes Place over Distance



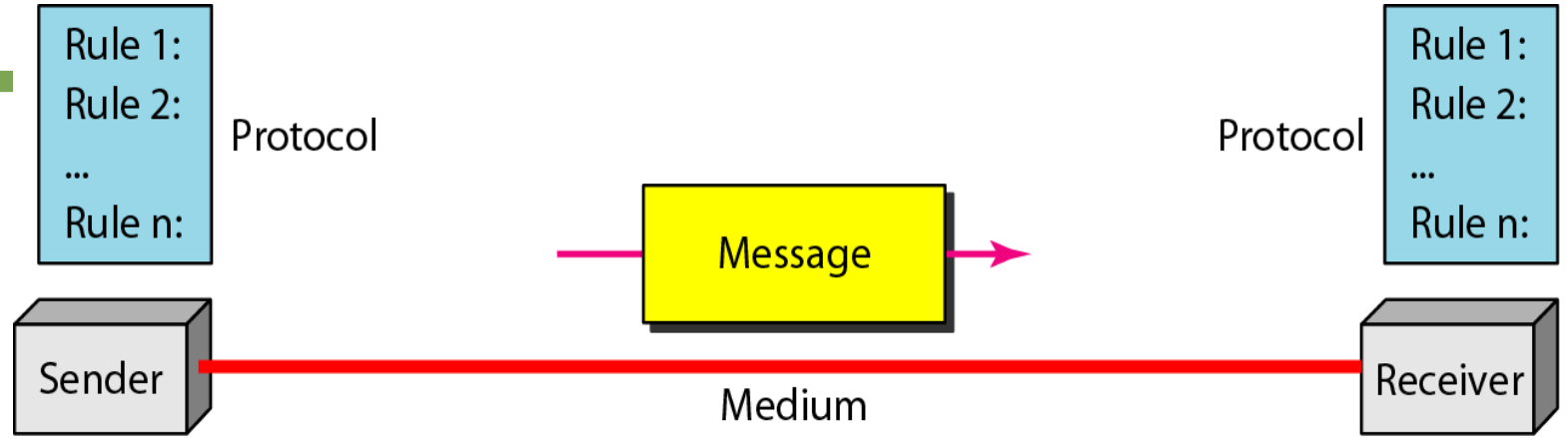
Data Communication

- The term telecommunication means communication at a distance.
- The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Characteristics of Data Communication

- **Delivery:** when a sender transmits data, it must reach only the intended destination
- **Accuracy:** The system must deliver the data accurately
- **Timeliness:** The system must deliver data without any delay.
- **Jitter:** Jitter refers to the variation in the packet arrival time.

Components of data communication



- **Message:** The message is the information (data) to be communicated. (Eg.) text, numbers, pictures, audio or video
- **Sender:** The sender is the device that sends the data message. It can be computer, telephone handset, video camera etc.,
- **Receiver:** The receiver is the device that receives the message. (Eg.) Computer, telephone handset, Television etc.,

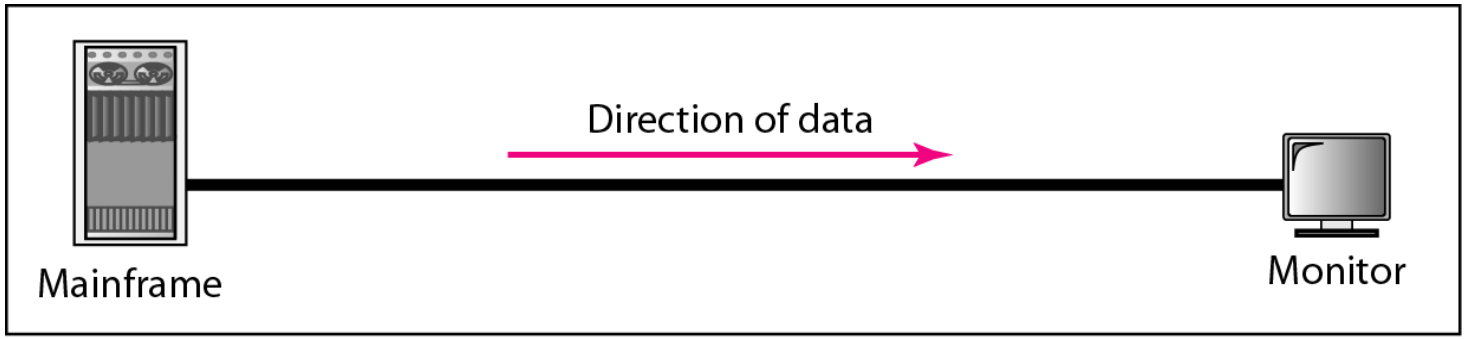
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- **Transmission Medium:** It is physical path by which a message travel from sender to receiver. (Eg) Twisted pair wire, coaxial cable, fiber optic cable or radio waves
- **Protocol:** A protocol is a set of rules that govern data communications.

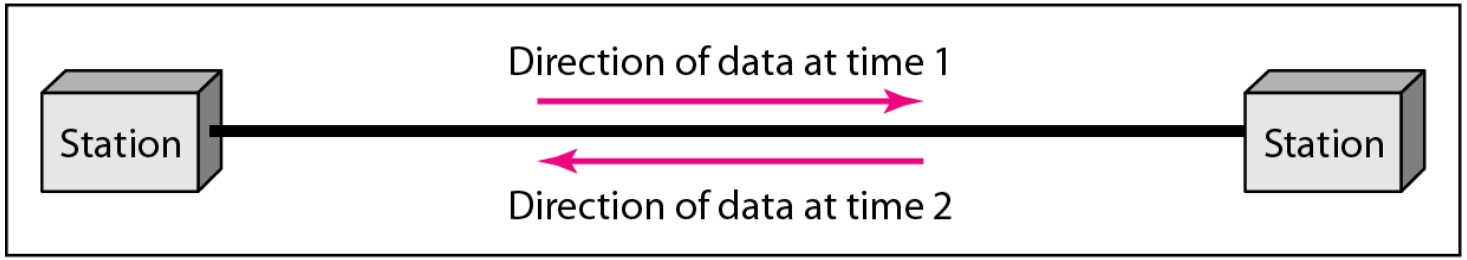
It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating (Ex) just as a person speaking French cannot be understood by a person who speaks only Japanese

Data Flow

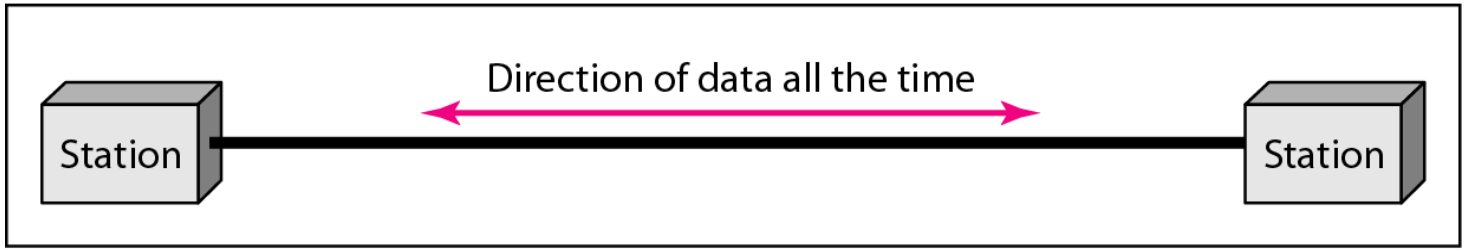
- Simplex
- Half-duplex
- Full-duplex



a. Simplex



b. Half-duplex



c. Full-duplex

Network Criteria

- **Performance:** Depends on Network Elements, Measured in terms of Delay and Throughput
- **Reliability:** Failure rate of network components, Measured in terms of availability/robustness
- **Security:** Data protection against corruption/loss of data due to
  - Errors
  - Malicious users

Physical Structures

- **Type of Connection**
  - Point to Point - single transmitter and receiver
  - Multipoint - multiple recipients of single transmission
- **Physical Topology**
  - Connection of devices
  - Type of transmission - unicast, mulitcast, broadcast

PHYSICAL STRUCTURES

1 TYPE OF CONNECTION

Point to Point
(single transmitter and receiver)



Data is sent from one device to one destination.

Multipoint
(multiple recipients of single transmission)



Data is sent from one device to multiple destinations.

2 PHYSICAL TOPOLOGY

Connection of Devices

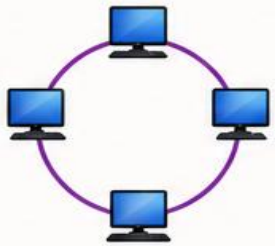
Bus Topology



Star Topology



Ring Topology



Physical layout showing how devices are connected in the network.

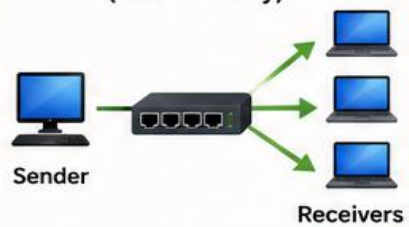
Type of Transmission

Unicast
(One to One)



Data is sent from one sender to one receiver.

Multicast
(One to Many)



Data is sent from one sender to a selected group of receivers.

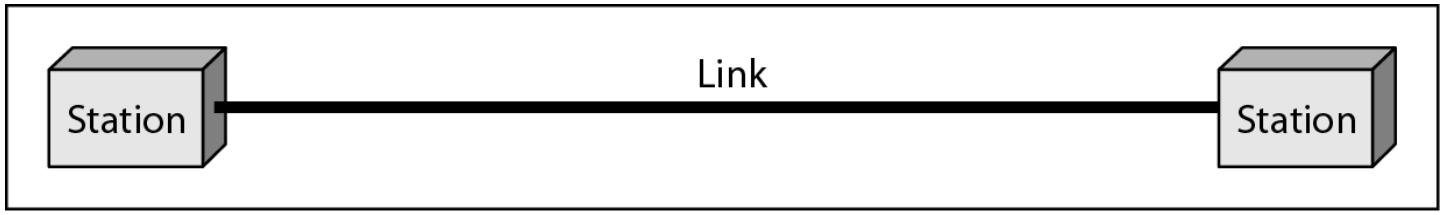
Broadcast
(One to All)



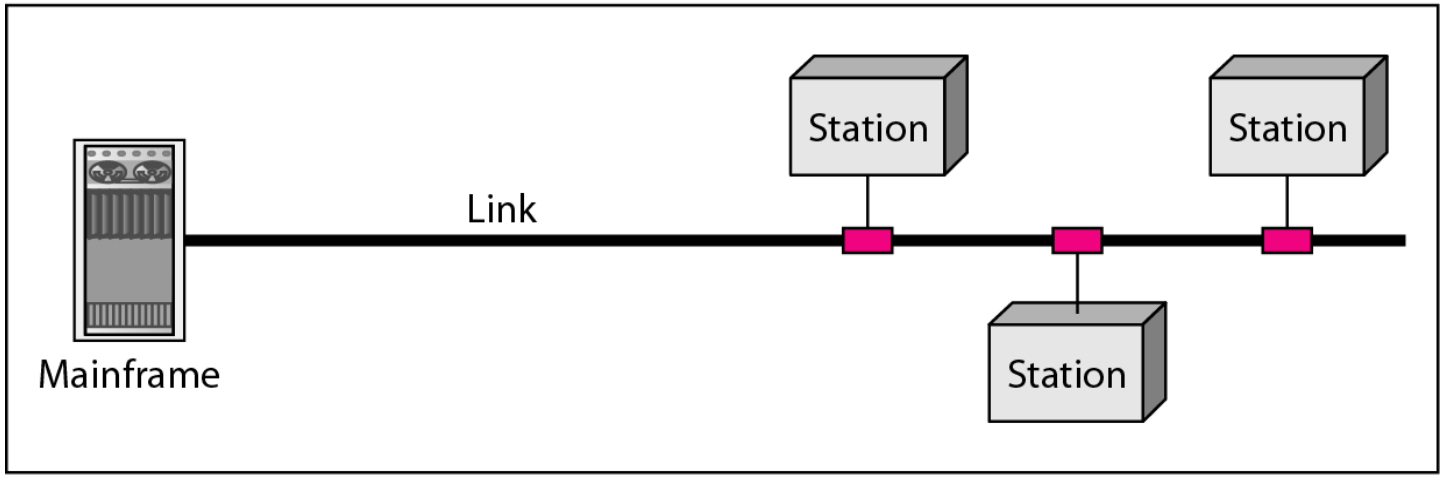
Data is sent from one sender to all devices on the network.

Physical structures define how devices are connected and how data is transmitted across a network.

Types of connections



a. Point-to-point



b. Multipoint

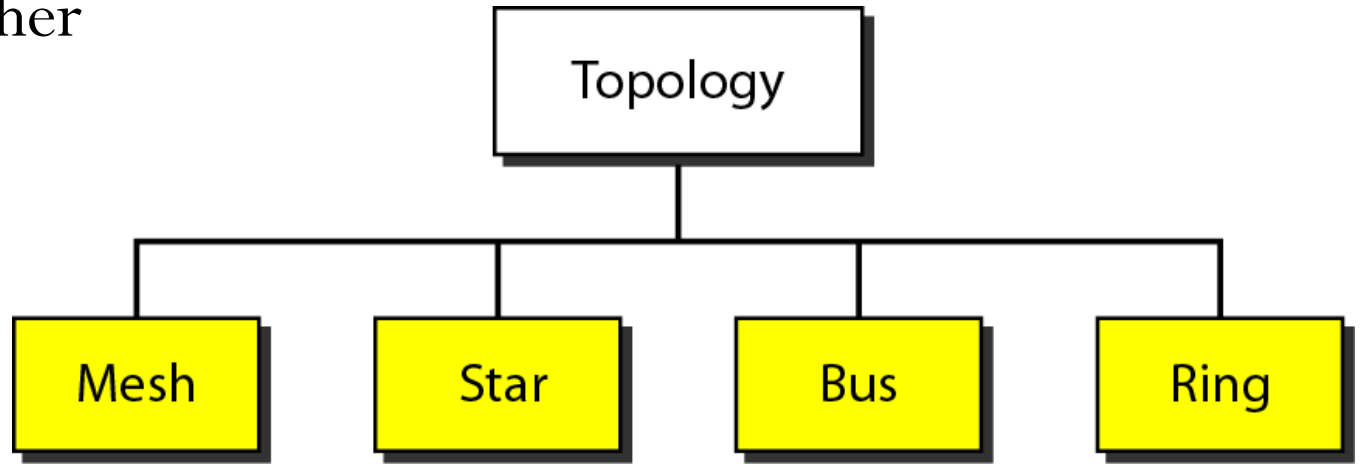
Networking Topologies
Network Architecture

Physical topology

- It refers to the way in which a network is laid out physically.
- Two or more devices connect to a link
- Two or more links form a topology

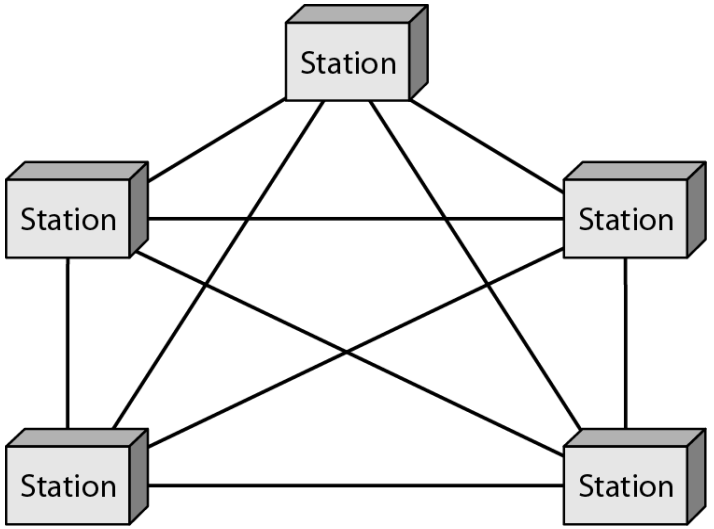
Topology:

It is the geometric representation of the relationship of all the links and linking devices(called nodes) to one another



A fully connected mesh topology (five devices)

- Each node is connected to every other node.
- Node 1 must be connected to n - 1 nodes, node 2 must be connected to n – 1 nodes, and finally node n must be connected to n - 1 nodes. We need n(n - 1) physical links.



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Advantages of Mesh topology

- Eliminating the traffic problems
- Robust (If one link becomes unusable, it does not incapacitate the entire system)
- privacy or security

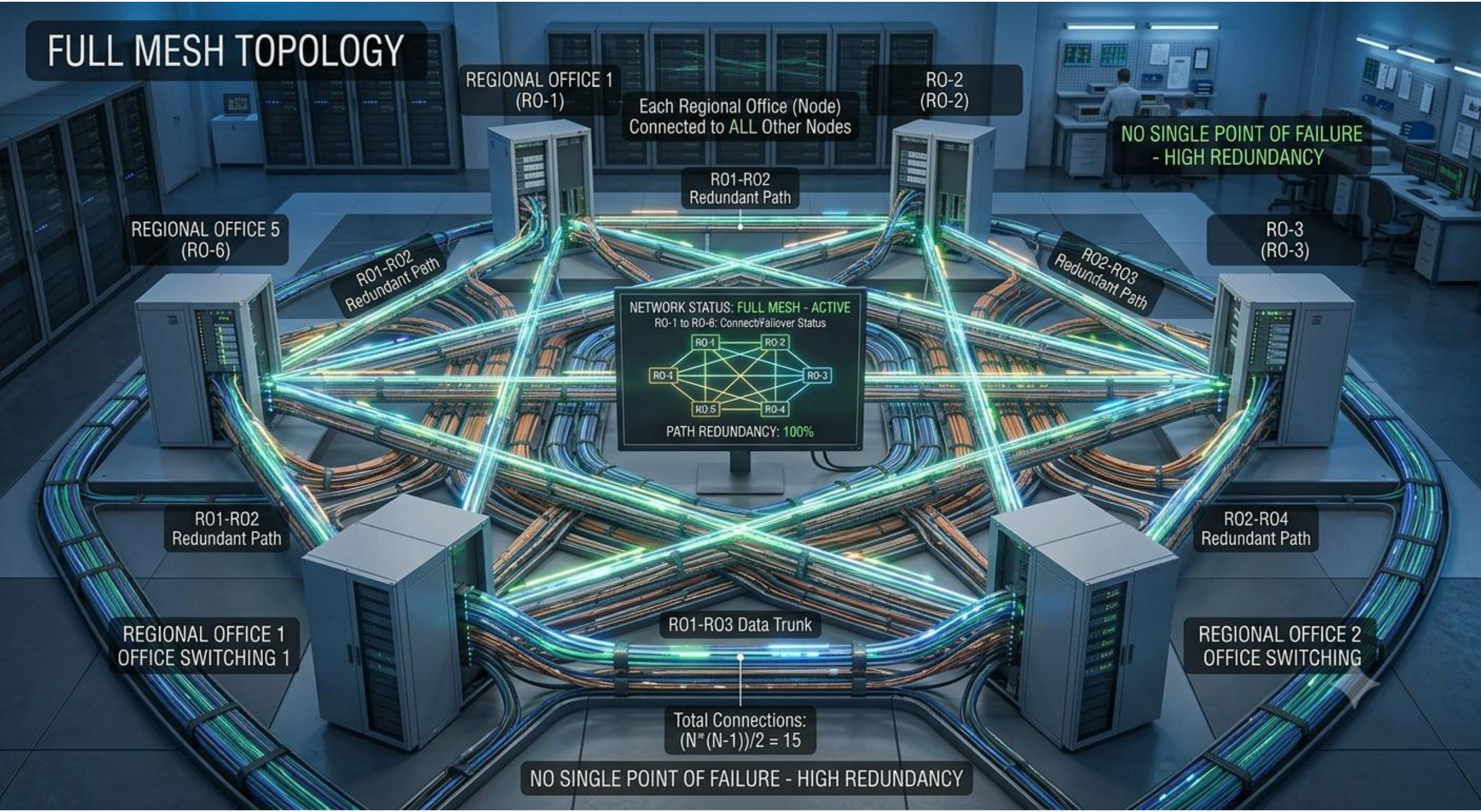
Disadvantages of Mesh Topology

- amount of cabling & number of I/O ports
- installation and reconnection are difficult
- Use of wires(in walls, ceilings, or floors)
- Hardware((I/O ports and cable) can be prohibitively expensive)

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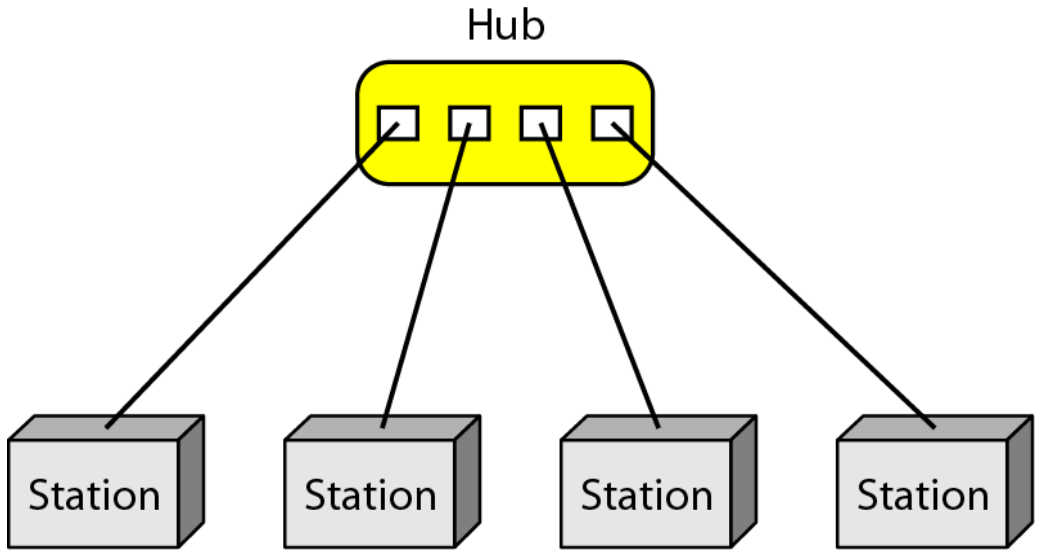
Practical example of a mesh topology

The connection of telephone regional offices in which each regional office needs to be connected to every other regional office.


FULL MESH TOPOLOGY
REGIONAL OFFICE 1
(RO-1)
Each Regional Office (Node)
Connected to ALL Other Nodes
RO-2
(RO-2)
NO SINGLE POINT OF FAILURE
- HIGH REDUNDANCY
RO1-RO2
Redundant Path
REGIONAL OFFICE 5
(RO-6)
RO1-RO2
Redundant Path
RO2-RO3
Redundant Path
RO-3
(RO-3)
NETWORK STATUS: FULL MESH - ACTIVE
RO-1 to RO-6: Connect/Faliover Status
RO-1
RO-2
RO-4
RO-3
RO-5
RO-4
PATH REDUNDANCY: 100%
RO1-RO2
Redundant Path
RO2-RO4
Redundant Path
REGIONAL OFFICE 1
OFFICE SWITCHING 1
RO1-RO3 Data Trunk
REGIONAL OFFICE 2
OFFICE SWITCHING
Total Connections:
(N*(N-1))/2 = 15
NO SINGLE POINT OF FAILURE - HIGH REDUNDANCY

A star topology connecting four stations

Each device has a dedicated point-to-point link only to a central controller, usually called a hub



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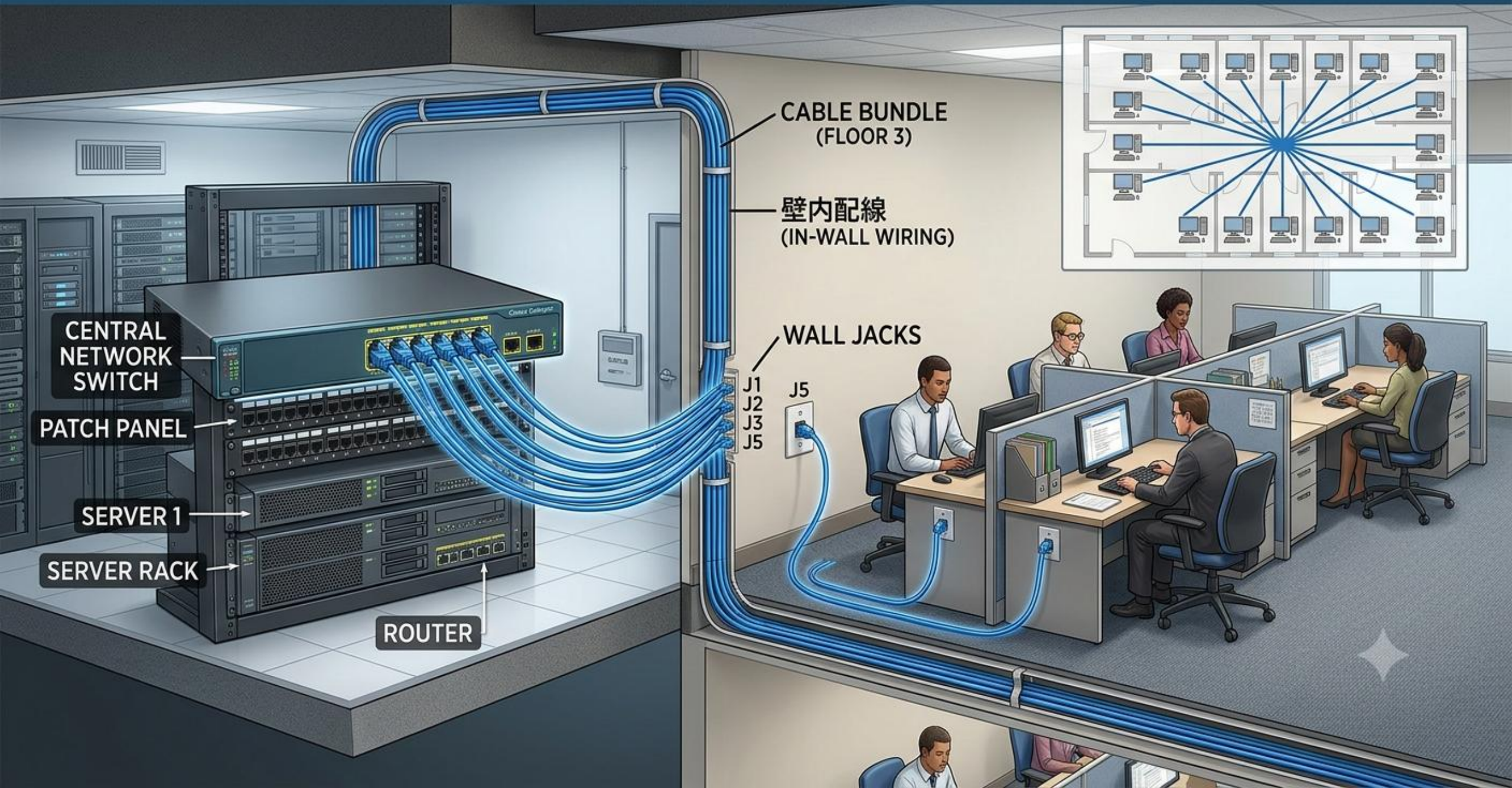
Advantages

- Less expensive than a mesh topology
- Easy to install and reconfigure
- Less cabling needs to be housed
- Robustness(If one link fails, only that link is affected. All other links remain active)

Disadvantage

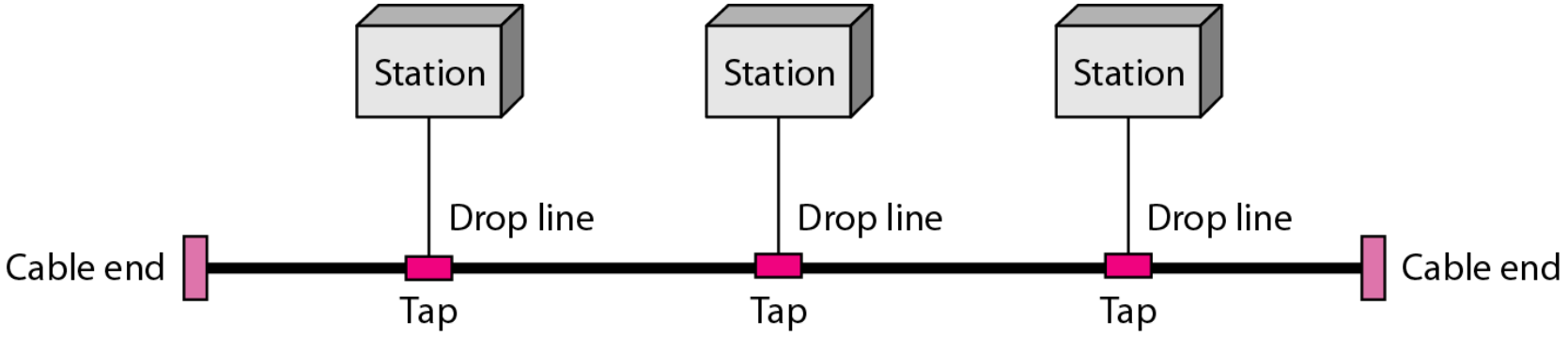
- Dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead
- Star topology is used in local-area networks (LANs)

STANDARD OFFICE LAN: STAR TOPOLOGY EXAMPLE



A bus topology connecting three stations

A bus topology is multipoint communication. One long cable acts as a backbone to link all the devices in a network



Contd...

- Nodes are connected to the bus cable by drop lines and taps.
- A drop line is a connection running between the device and the main cable.
- A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.
- As a signal travels along the backbone, some of its energy is transformed into heat.
- Therefore, it becomes weaker and weaker as it travels farther and farther.
- For this reason there is a limit on the number of taps a bus can support and on the distance between those taps

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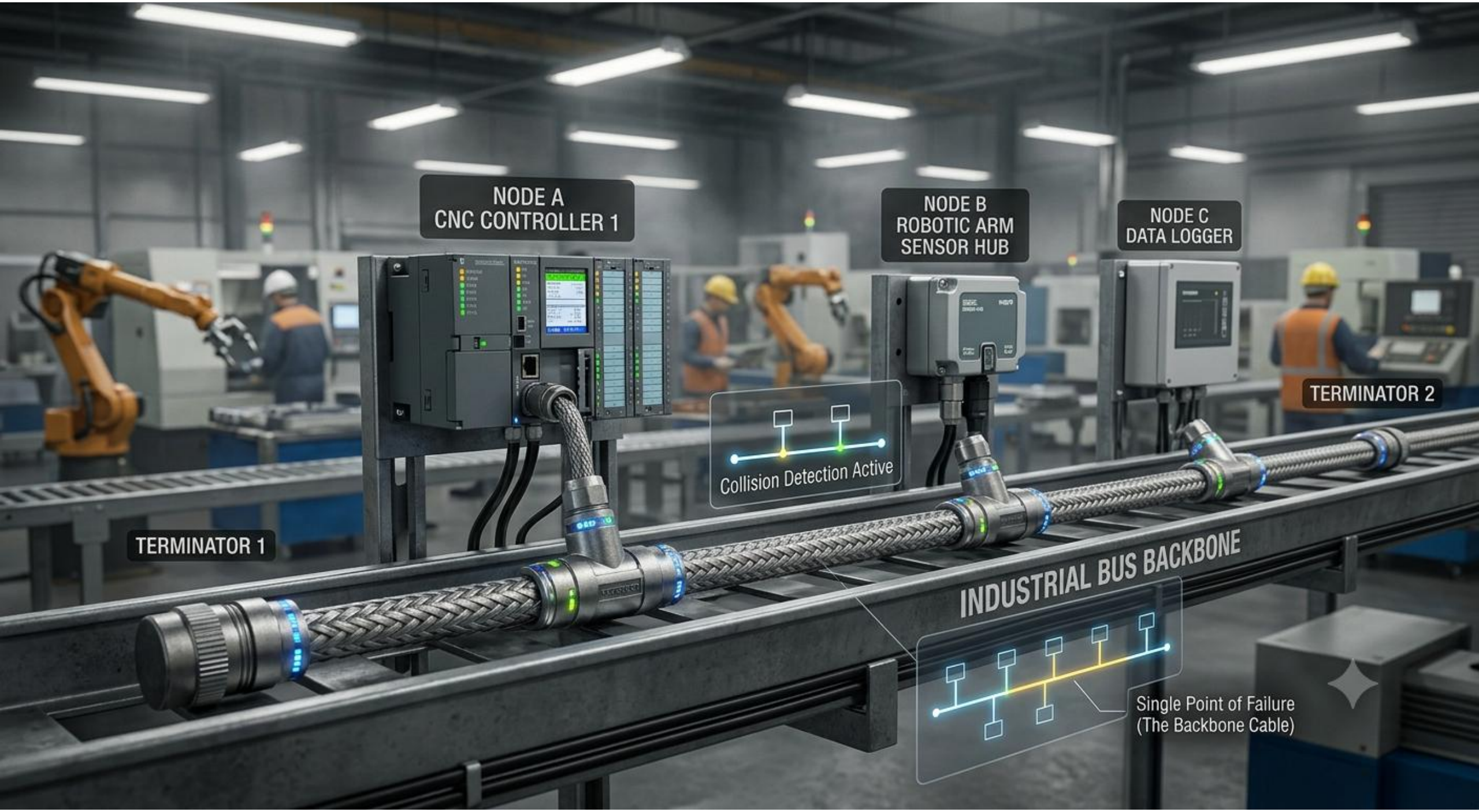
Advantages:

- Bus topology include ease of installation.
- Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths.
- A bus uses less cabling than mesh or star topologies.
- In a star, for example, four network devices in the same room require four lengths of cable reaching all the way to the hub. In a bus, this redundancy is eliminated.
- Only the backbone cable stretches through the entire facility. Each drop line has to reach only as far as the nearest point on the backbone.

Contd...

Disadvantages

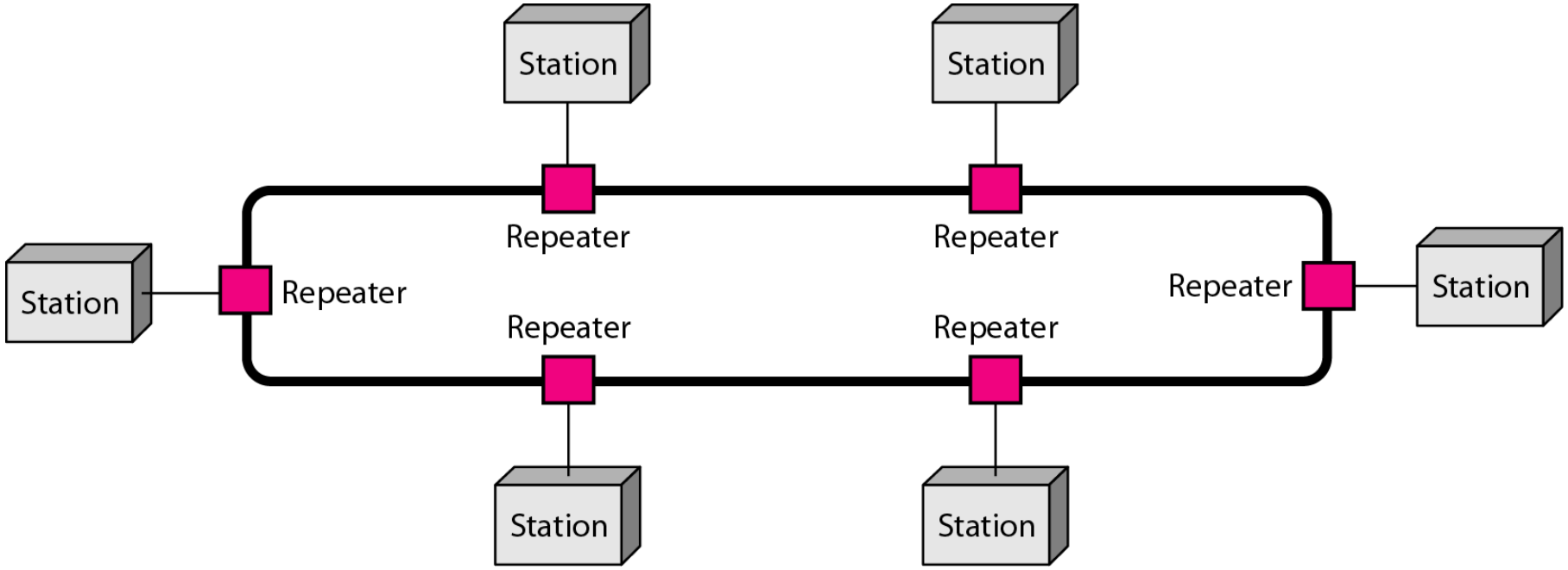
- Difficult reconnection and fault isolation.
- A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices.
- Signal reflection at the taps can cause degradation in quality.
- This degradation can be controlled by limiting the number and spacing of devices connected to a given length of cable.
- Adding new devices may therefore require modification or replacement of the backbone.


NODE A
CNC CONTROLLER 1
NODE B
ROBOTIC ARM
SENSOR HUB
NODE C
DATA LOGGER
TERMINATOR 2
Collision Detection Active
TERMINATOR 1
INDUSTRIAL BUS BACKBONE
Single Point of Failure
(The Backbone Cable)

Ring Topology

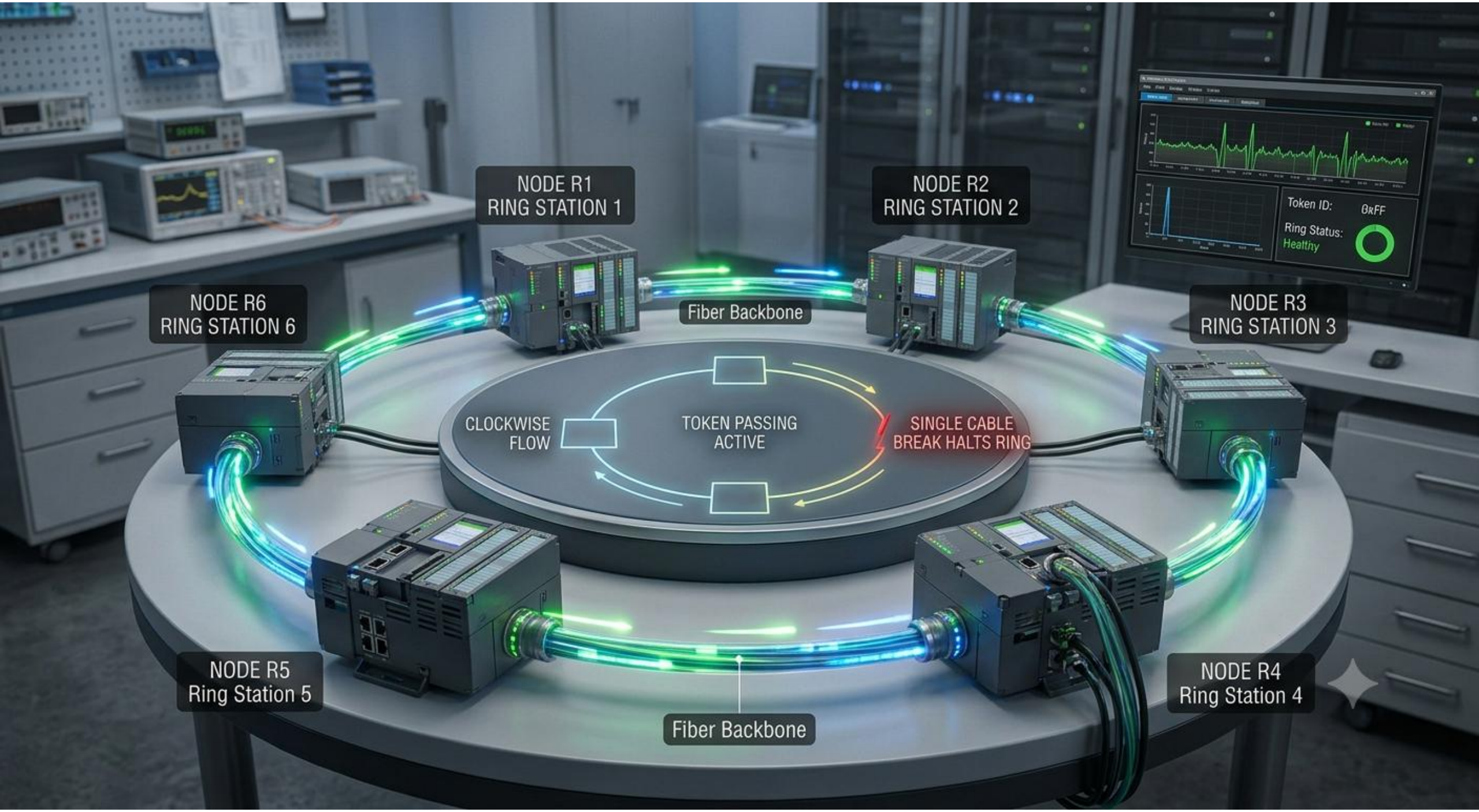
- In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater.
- When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.

A ring topology connecting six stations



Contd...

- A ring is relatively easy to install and reconfigure.
- Each device is linked to only its immediate neighbors (either physically or logically).
- To add or delete a device requires changing only two connections.
- The only constraints are media and traffic considerations (maximum ring length and number of devices).
- In addition, fault isolation is simplified. Generally, in a ring a signal is circulating at all times.
- If one device does not receive a signal within a specified period, it can issue an alarm.


NODE R1
RING STATION 1
NODE R2
RING STATION 2
NODE R6
RING STATION 6
NODE R3
RING STATION 3
Fiber Backbone
CLOCKWISE
FLOW
TOKEN PASSING
ACTIVE
SINGLE CABLE
BREAK HALTS RING
Token ID:
0xFF
Ring Status:
Healthy
NODE R5
Ring Station 5
NODE R4
Ring Station 4
Fiber Backbone

Class Poll

- Difficult reconnection and fault isolation are the disadvantages of
  A. Star Topology
  B. Mesh Topology
  C. Bus Topology
  D. Ring Topology

Categories of Networks

- Local Area Networks (LANs)
- Wide Area Networks (WANs)
- Metropolitan Area Networks (MANs)

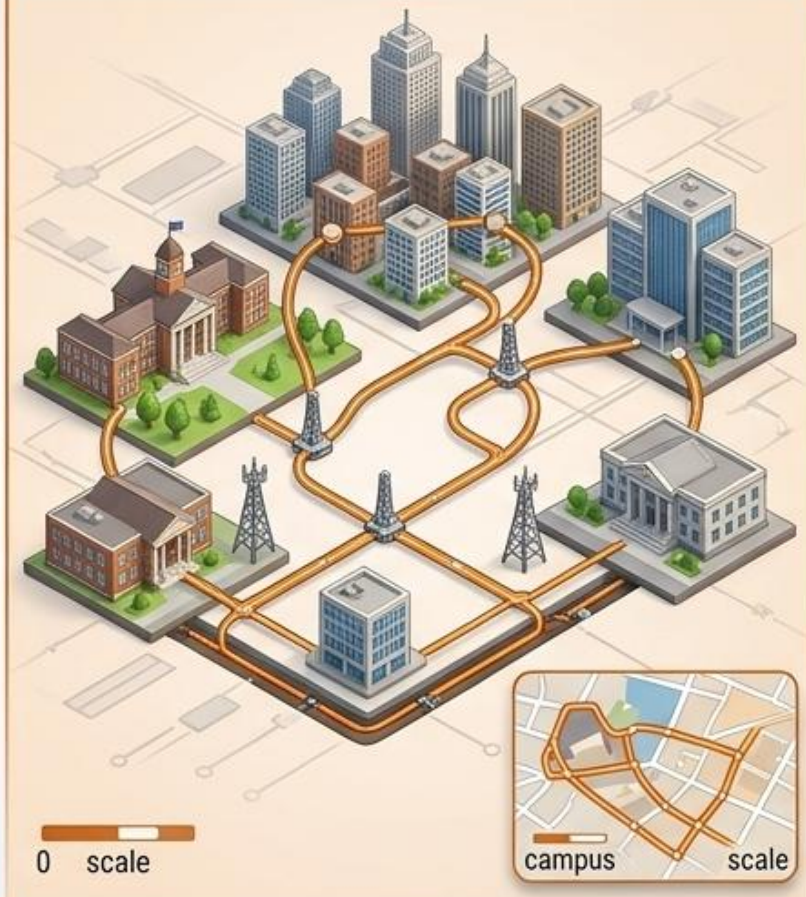
COMPARISON OF NETWORK TYPES

Local Area Networks (LANs)



- **Local Area Networks (LANs)**
  - Short distances
  - Designed to provide local interconnectivity

Metropolitan Area Networks (MANs)



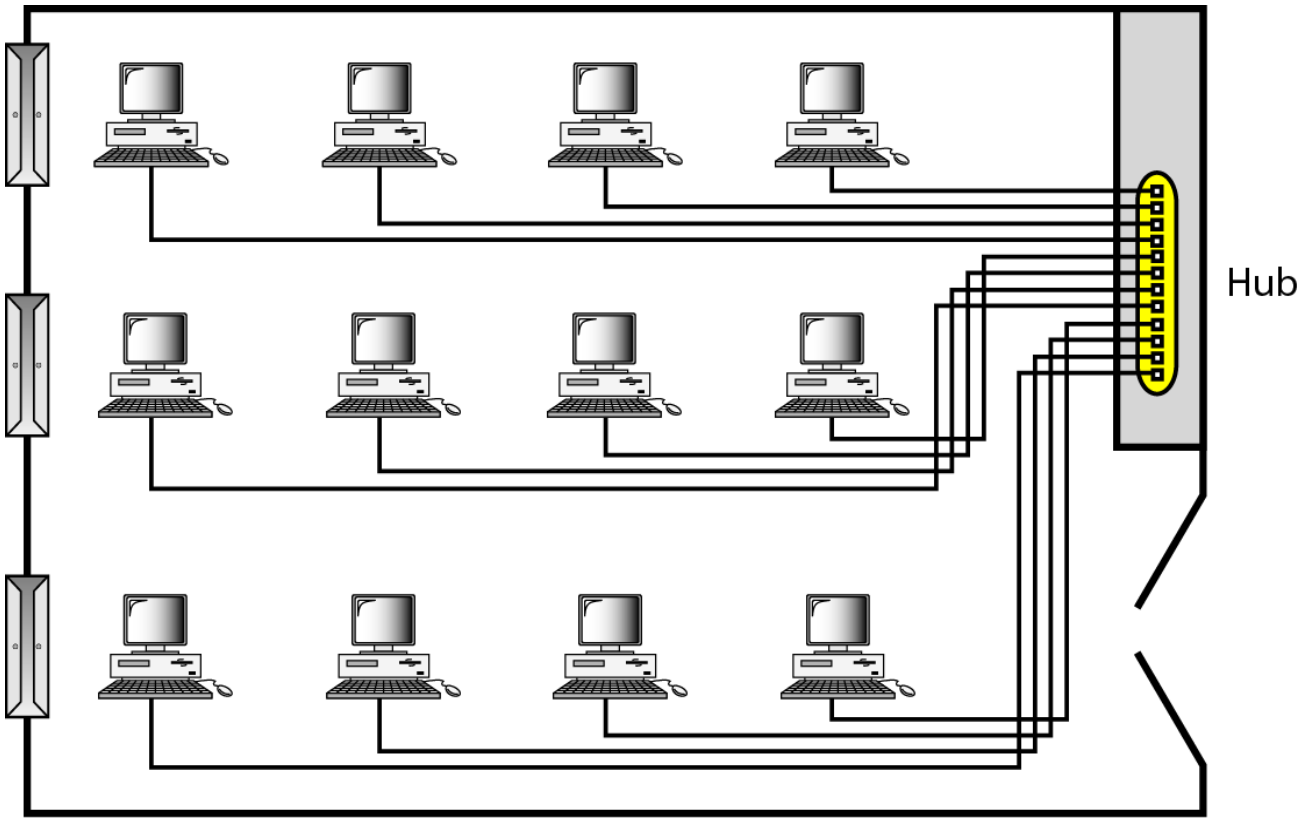
- **Metropolitan Area Networks (MANs)**
  - Provide connectivity over areas such as a city, a campus

Wide Area Networks (WANs)



- **Wide Area Networks (WANs)**
  - Long distances
  - Provide connectivity over large areas

An isolated LAN connecting 12 computers to a hub in a closet



Contd...

- A local area network (LAN) is usually privately owned and links the devices in a single office, building, or campus
- Depending on the needs of an organization and the type of technology used, a LAN can be as simple as two PCs and a printer in someone's home office;
- Currently, LAN size is limited to a few kilometers.
- LAN will use only one type of transmission medium.
- The most common LAN topologies are bus, ring, and star.
- Early LANs had data rates in the 4 to 16 megabits per second (Mbps) range.
- Speeds are normally 100 or 1000 Mbps.
- Wireless LANs are the newest evolution in LAN technology

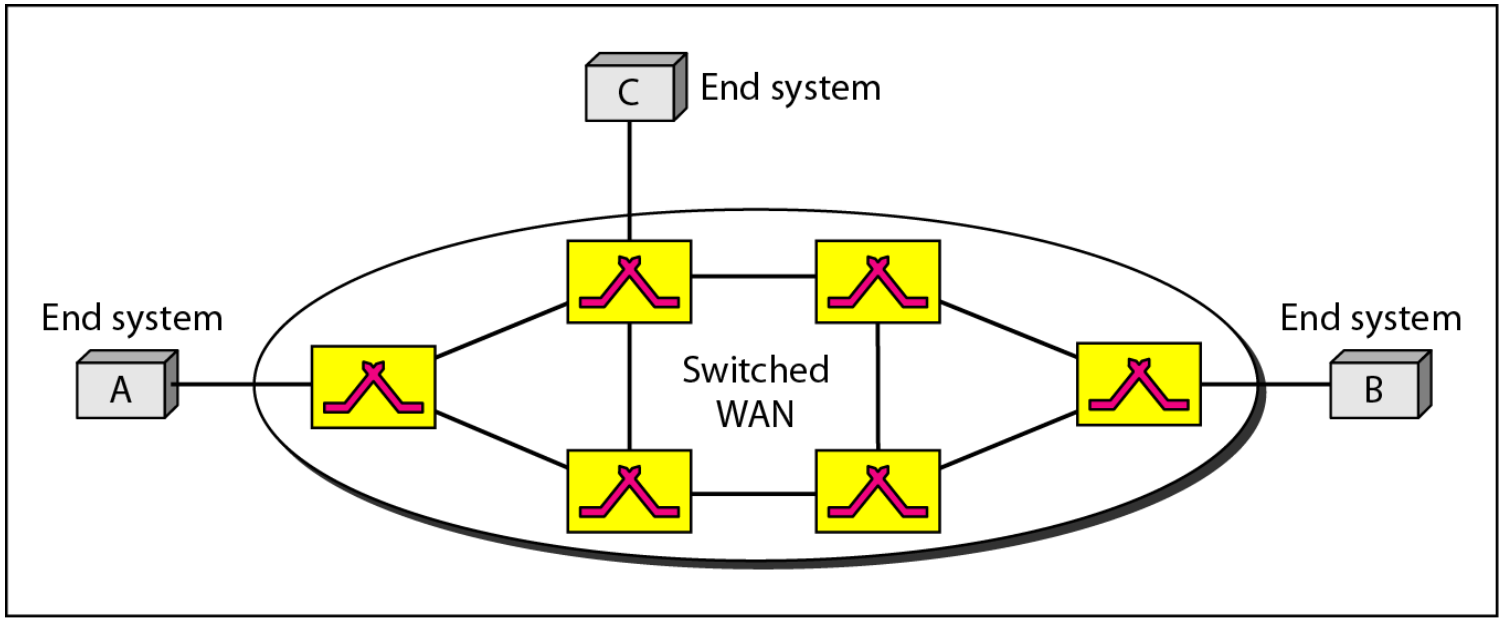
Wide Area Network (WAN)

- A wide area network (WAN) provides **long-distance transmission of data, image, audio, and video information over large geographic areas** that may comprise a country, a continent, or even the whole world.
- A WAN can be as complex as the backbones that connect the Internet or as simple as a dial-up line that connects a home computer to the Internet.

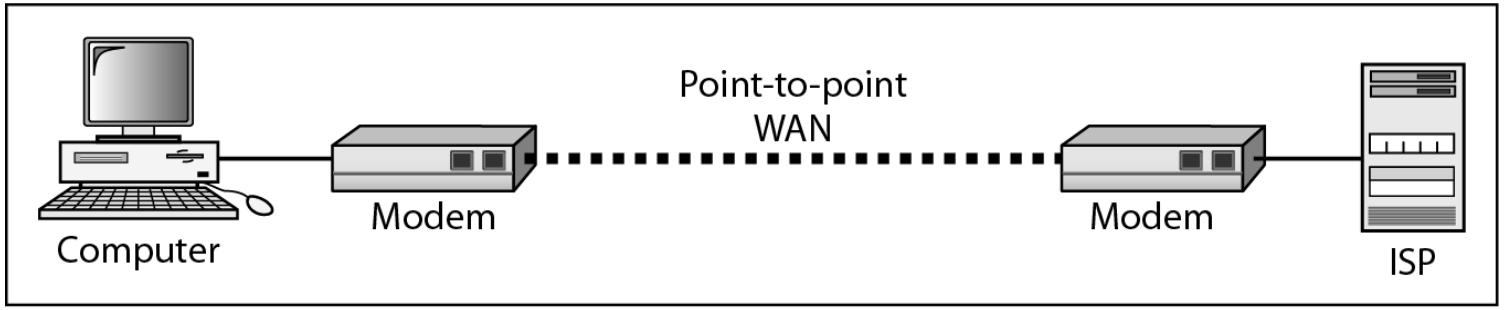
Two types of WAN

- switched WAN
- point-to-point WAN

WANs: a switched WAN and a point-to-point WAN



a. Switched WAN



b. Point-to-point WAN

TYPES OF WAN: EXAMPLES

1. POINT-TO-POINT WAN
(Dedicated Connection)

- A dedicated link is established between two locations. Only these two sites can communicate.



REAL-LIFE EXAMPLE



Bank Headquarters ↔ Branch Office

- Private & Dedicated Connection
- High Security
- Guaranteed Bandwidth
- Used for two locations only

TECHNOLOGIES


Leased Line


Metro Ethernet


Optical Fiber (P2P)

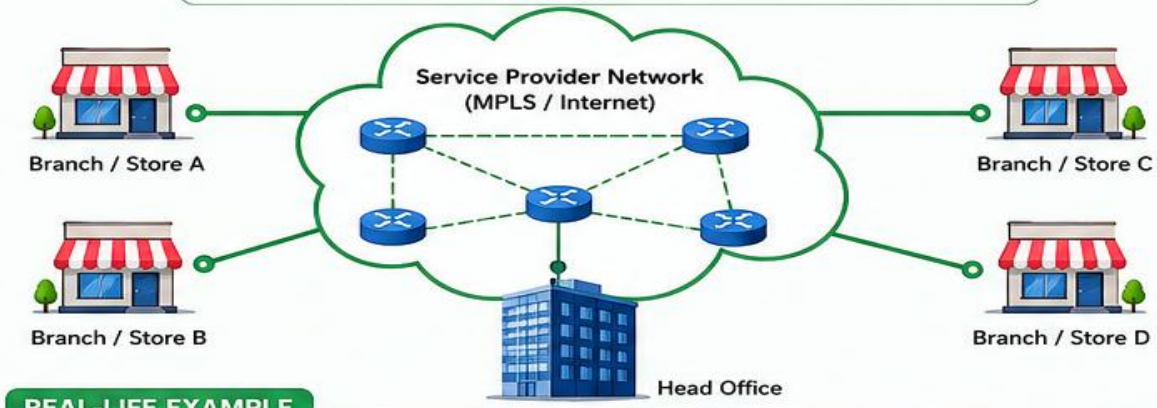

MPLS Private Circuit

COMMON EXAMPLES

- Bank Head Office ↔ Branch
- University Main Campus ↔ Research Center
- Company HQ ↔ Manufacturing Plant
- ATM Network ↔ Bank Data Center

2. SWITCHED WAN
(Shared Connection)

- Multiple locations connect through a service provider's switched network. Resources are shared.



REAL-LIFE EXAMPLE



Retail Chain with Multiple Stores

- Cost Effective
- Scalable – Easy to add more sites
- Bandwidth Shared
- Data is routed through the provider network

TECHNOLOGIES


MPLS


VPN over Internet


SD-WAN


Frame Relay (Legacy)


ATM (Legacy)

COMMON EXAMPLES

- Retail Stores connected via MPLS
- Hospital Branches connected via MPLS
- University Campuses connected via ISP
- Company Offices connected via Internet VPN

QUICK COMPARISON

	Connection	Number of Sites	Cost	Performance	Scalability	Example
Point-to-Point WAN	Dedicated (One-to-One)	Two	Higher	Guaranteed	Limited	Bank HQ ↔ Branch
Switched WAN	Shared (Many-to-Many)	Multiple	Lower	Shared Bandwidth	Highly Scalable	Retail Chain Stores

EASY ANALOGY



Point-to-Point WAN: Like taking a private taxi directly from your home to your office.



Switched WAN: Like using the city road network, where many vehicles share the roads to reach different destinations.

Contd...

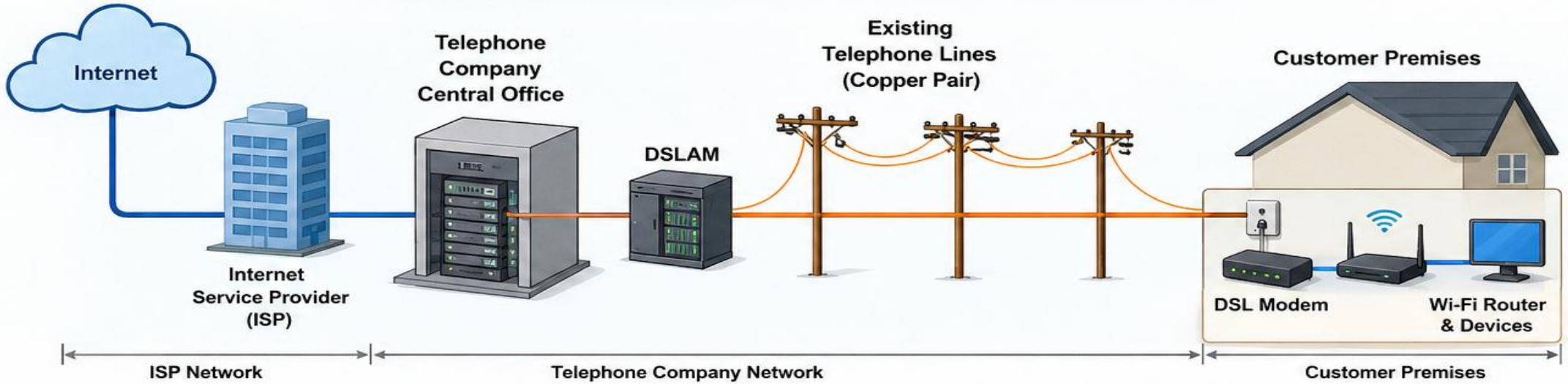
- The Switched WAN connects the end systems, which usually comprise a router (internetworking connecting device) that connects to another LAN or WAN.
- The Point-to-Point WAN is normally a line leased from a telephone or cable TV provider that connects a home computer or a small LAN to an Internet service provider (lSP).
- This type of WAN is often used to provide Internet access.

Metropolitan Area Network (MAN)

- A metropolitan area network (MAN) is a **network with a size between a LAN and a WAN.**
- It normally **covers the area inside a town or a city.**
- It is **designed for customers who need a high-speed connectivity**, normally to the Internet, and have endpoints spread over a city or part of city.

Example 1: Telephone Company Network (DSL)

Provides High-Speed DSL Line to the Customer



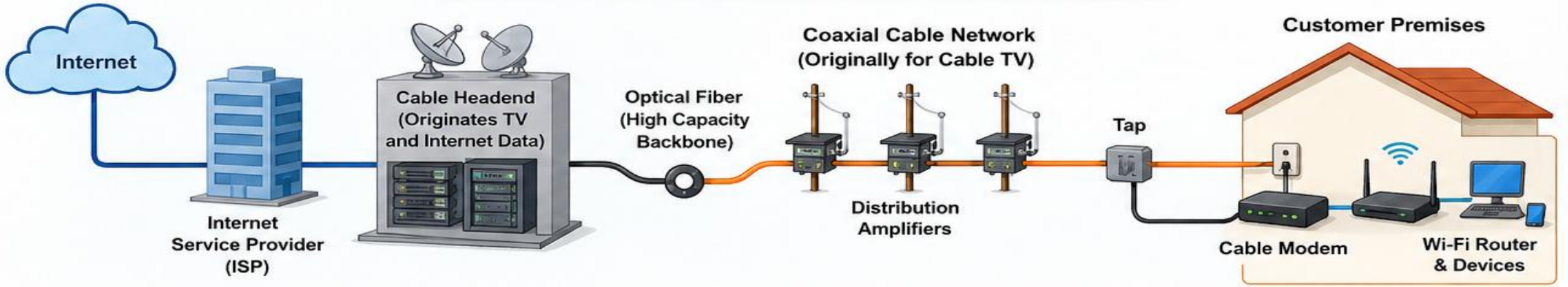
How it works

- DSLAM sends data over existing telephone lines.
- DSL Modem at customer end converts the signal and provides high-speed Internet access.

High-Speed DSL Internet

Example 2: Cable TV Network

Originally for Cable TV, Now Used for High-Speed Internet



How it works

- Data travels over the same coaxial cables used for cable TV.
- Cable Modem at customer end converts the signal and provides high-speed Internet.

High-Speed Internet Over Cable

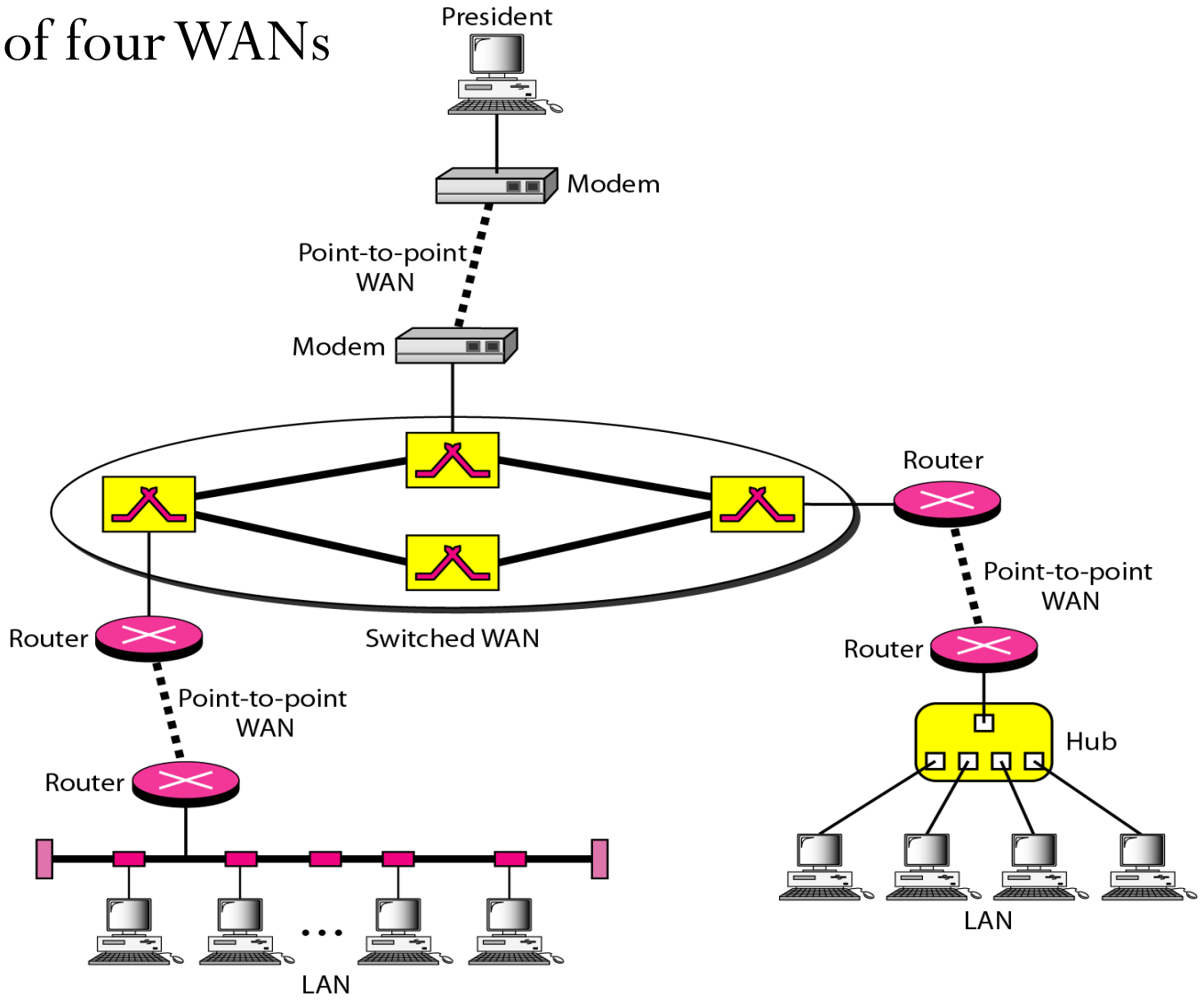
Both use existing infrastructure
No need for new dedicated lines to every home.

High-Speed Internet Access
Delivered efficiently to millions of customers.

Dual Purpose Networks
Telephone lines for voice + data (DSL)
Coaxial cables for TV + Internet (Cable)

Internetwork: Interconnection of Networks

A heterogeneous network made of four WANs and two LANs



Protocol

- A protocol is synonymous with rule. It consists of a set of rules that govern data communications.
- It determines what is communicated, how it is communicated and when it is communicated.
- The key elements of a protocol are syntax, semantics and timing

Elements of a Protocol

- Syntax
  - Structure or format of the data
  - Indicates how to read the bits
- Semantics
  - Interprets the meaning of the bits
  - Knows which fields define what action
- Timing
  - When data should be sent and what
  - Speed at which data should be sent or speed at which it is being received.

Network Architecture

- A computer network must provide general, cost effective and robust connectivity among a large number of computers.
- To deal with this complexity network designers developed the concept of network architecture which introduces a layered protocols or functions that guides the design and implementation of networks

Most widely network architectures are

- OSI Architecture
- Internet Architecture

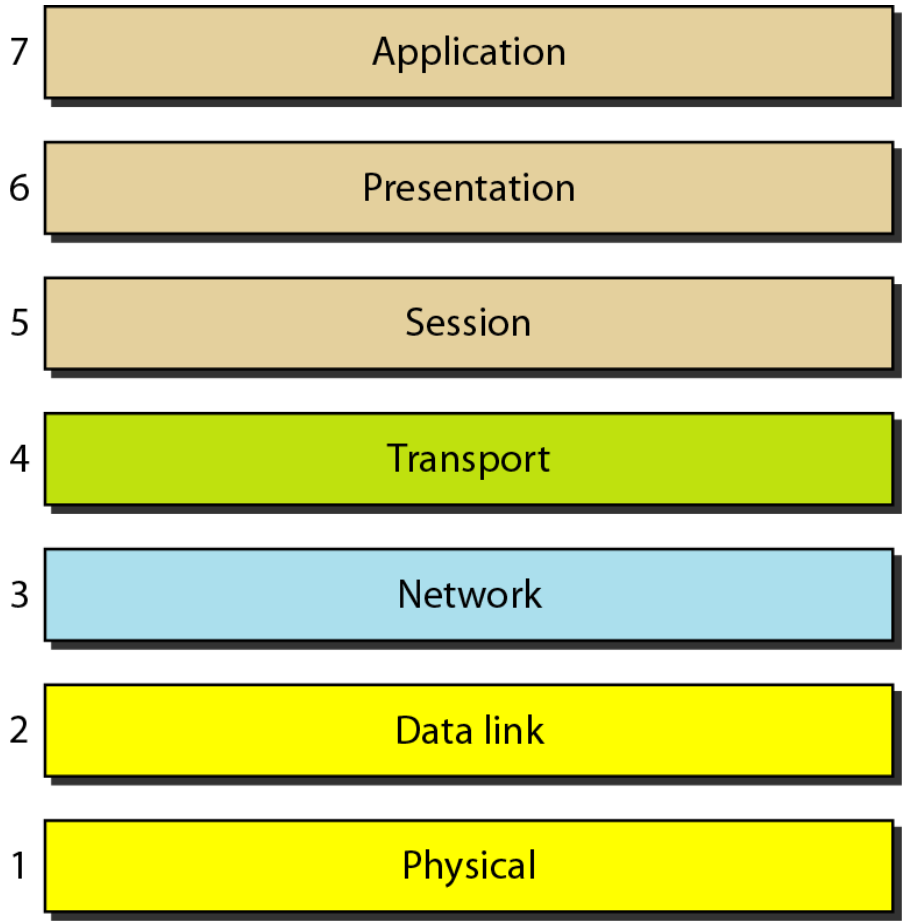
Network Model – OSI Model

- An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s

Note

ISO is the organization.
OSI is the model.

Contd...


7
Application
6
Presentation
5
Session
4
Transport
3
Network
2
Data link
1
Physical

Contd...

Seven layers can be split into three subgroups

Layer 1,2,3 : Network support layers

- It deals with **physical aspects of moving data from one device to another** (electrical specifications, physical connection, physical addressing, timing and reliability)

Layer 5,6&7: (implemented in software) - User support layers

- It allows **interoperability among unrelated software systems**

Layer 4

- It links the two subgroups (lower layers and upper layer)
- It ensures **end-to-end reliable data transmission**

OSI MODEL & NETWORK DEVICE CLASSIFICATION

Software Layers (Lower Layers)

7 Layer 7: **Application** - End User Apps (Browser, Email)

6 Layer 6: **Presentation**

5 Layer 5: **Session**

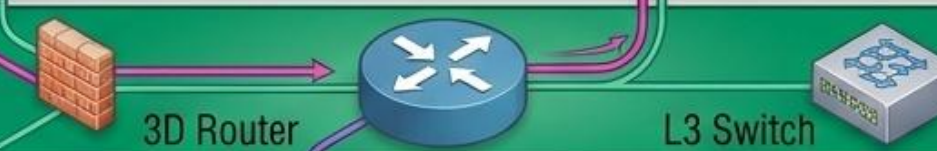
Network & Hardware Layers

4 Layer 4: **Transport**



Key Devices: Transort, Transport, Switches, and Firetoads
Protocols: IP, ICMP, Routing Protocols

3 Layer 3: **Network**



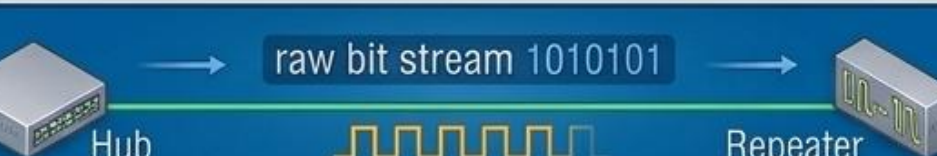
KEY DEVICE: ROUTERS, L3 SWITCHES.
Function: Path determination (Routing), Logical addressing (IP).
Protocols: IP, ICMP

2 Layer 2: **Data Link**



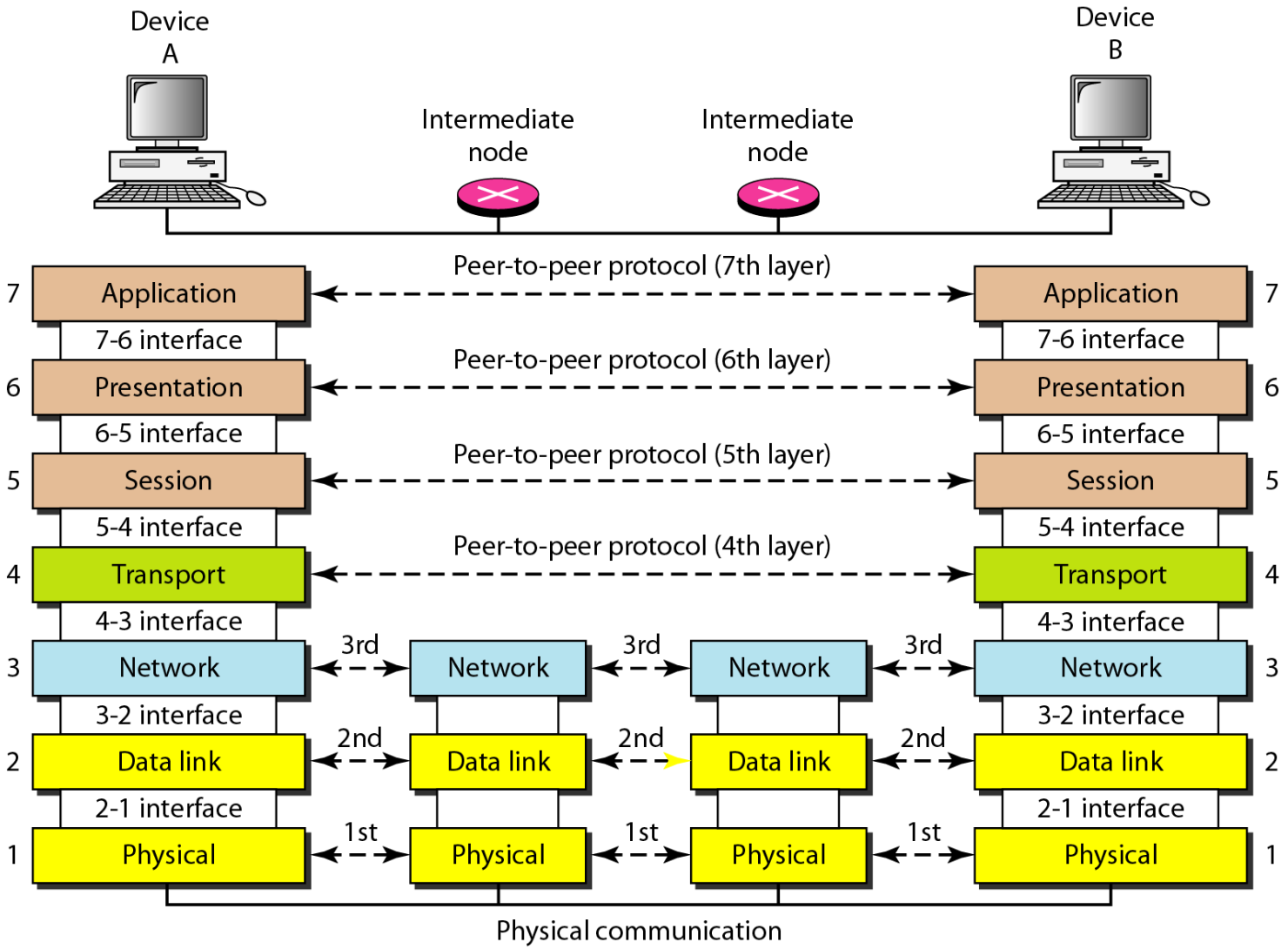
KEY DEVICES: SWITCHES, BRIDGES.
Function: Physical addressing (MAC),
Action: Reads MAC addresses.
Protocols: Ethernet, ARP

1 Layer 1: **Physical**

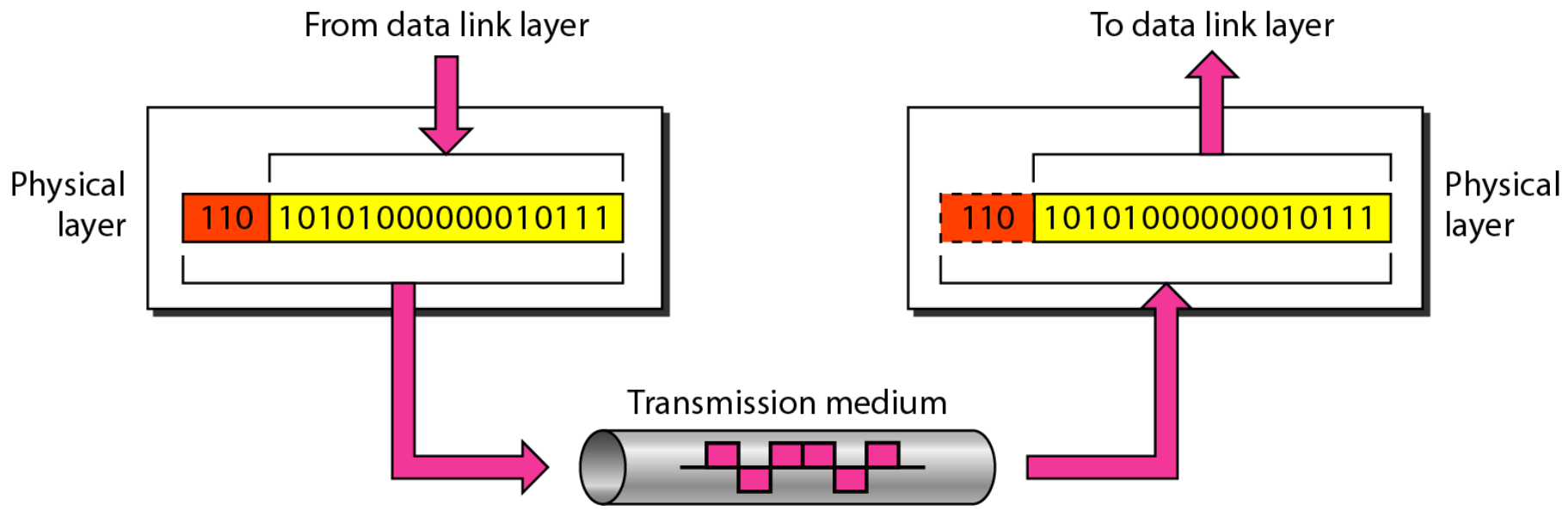


KEY DEVICES: HUBS, REPEATERS.
Function: Bits transmission, Electrical/ Physical connections.
Action: Broadcasts bits (no addressing).
Media: Cables, Wireless

Interaction between layers in the OSI model

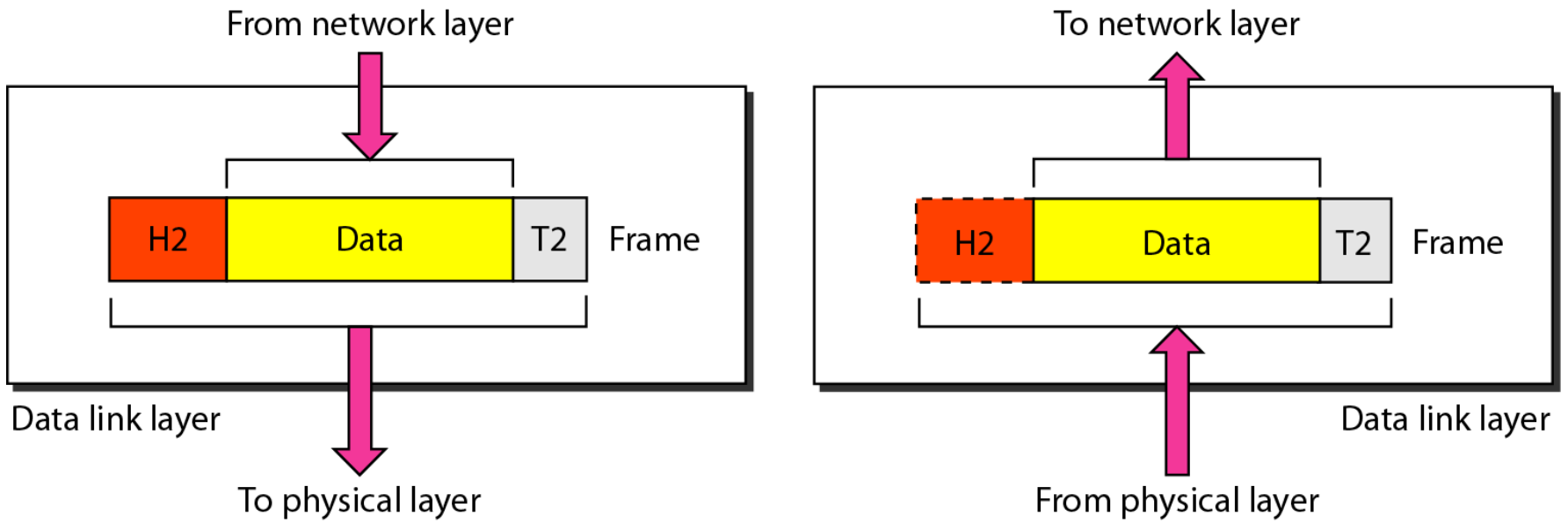


Physical Layer



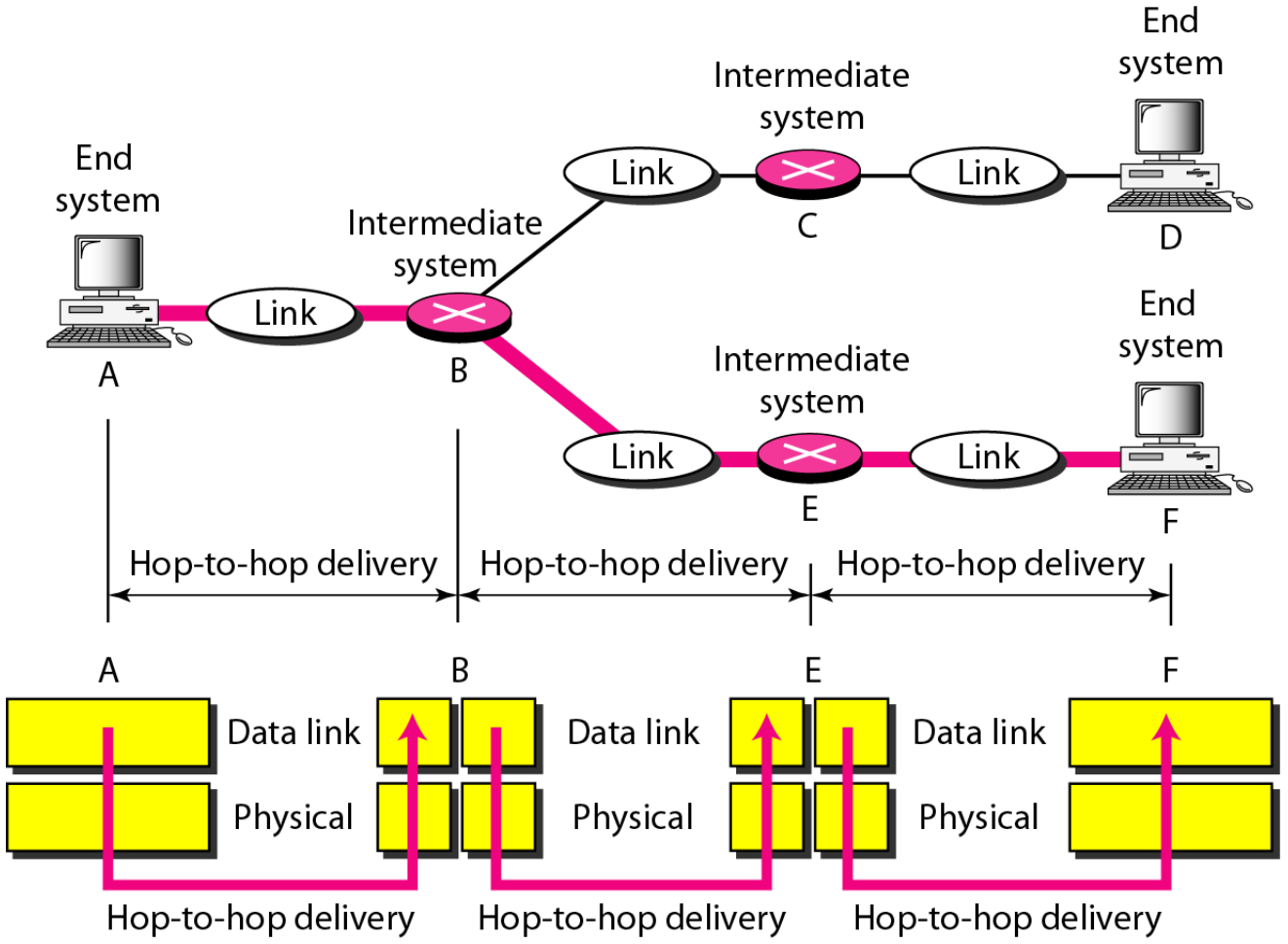
The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Data Link Layer

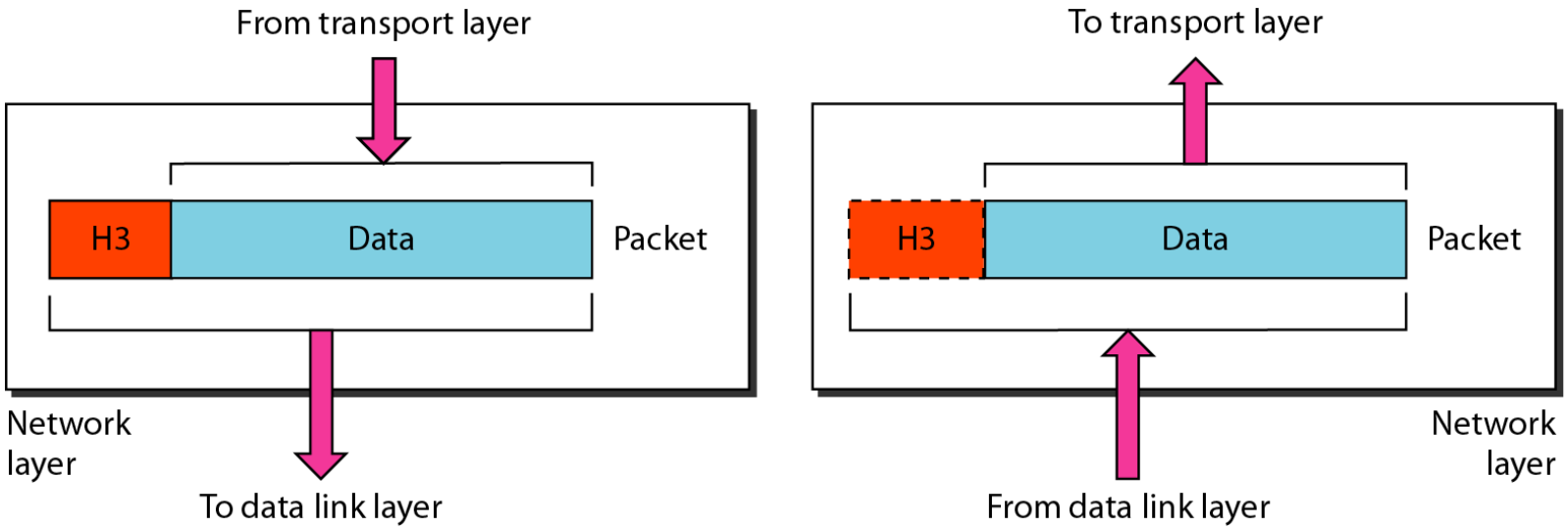


The data link layer is responsible for moving frames from one hop (node) to the next.

Hop-to-hop delivery

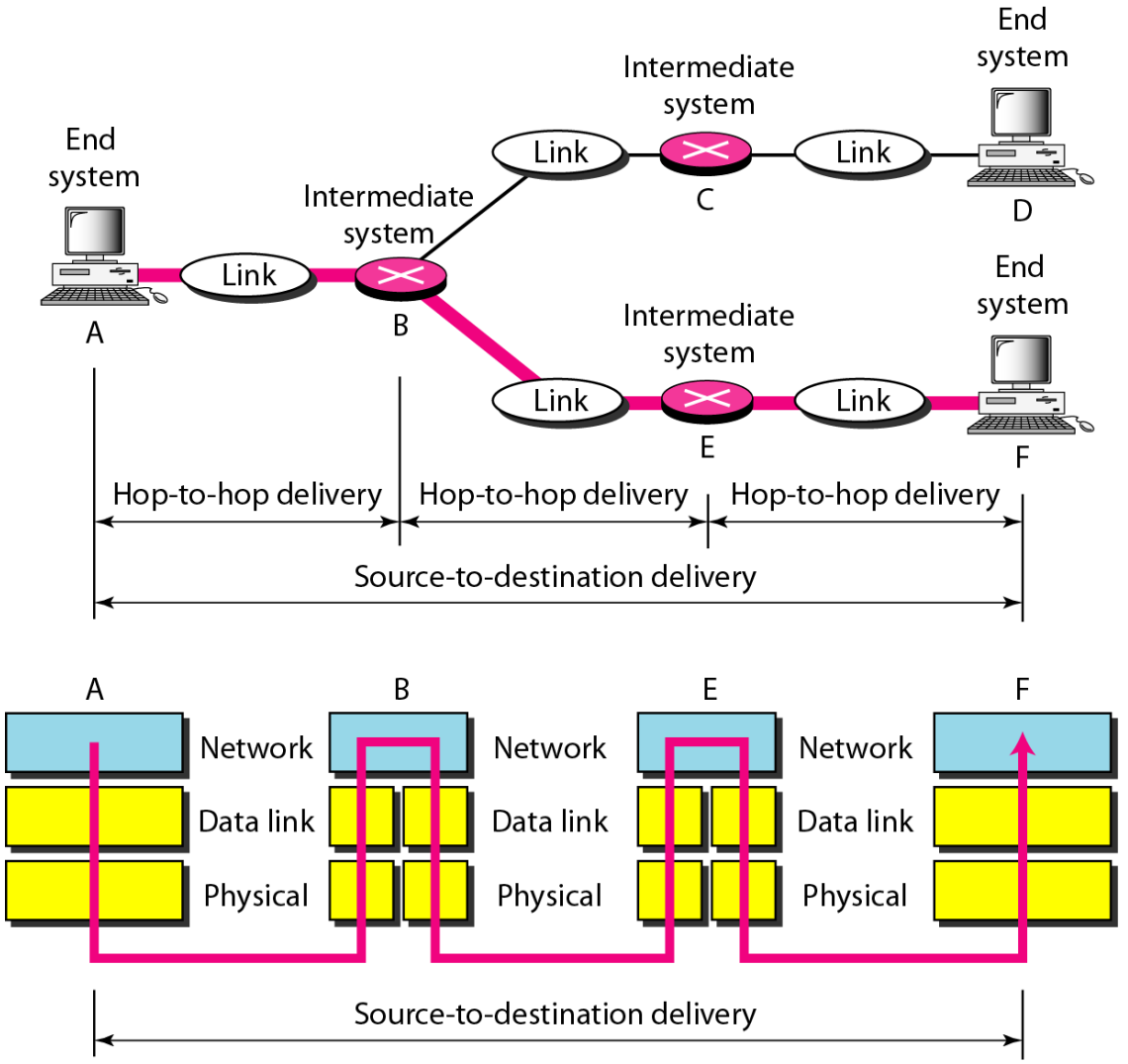


Network Layer

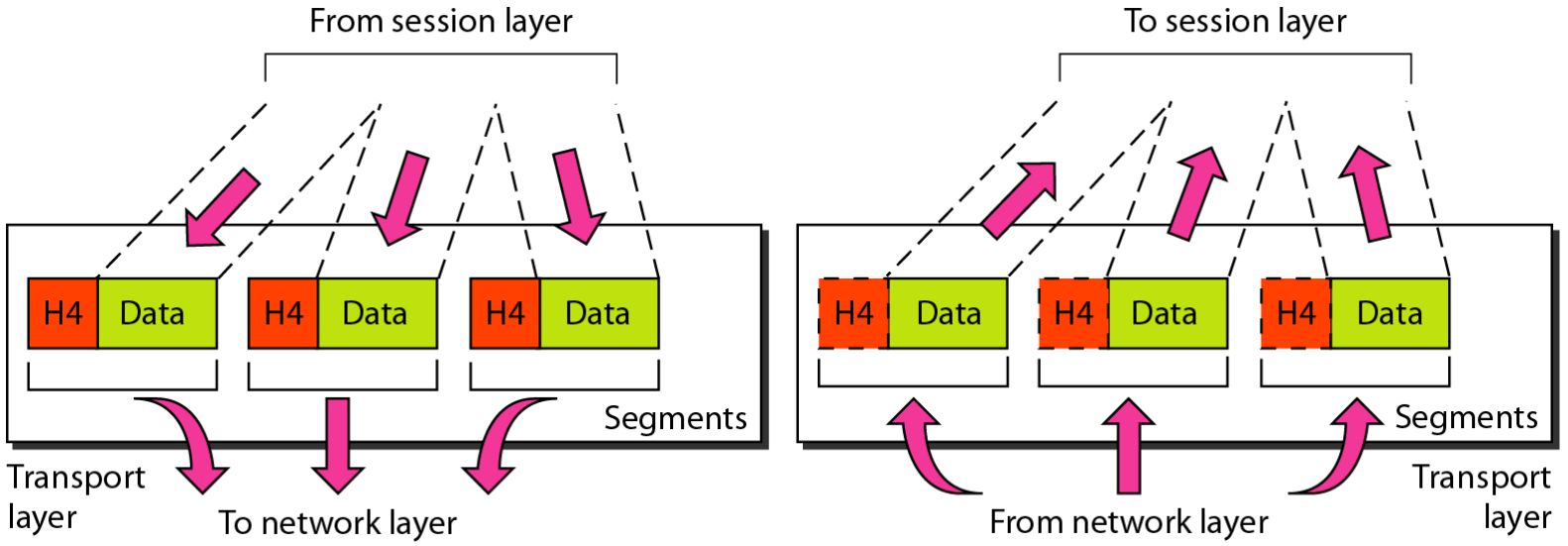


The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Source-to-destination delivery

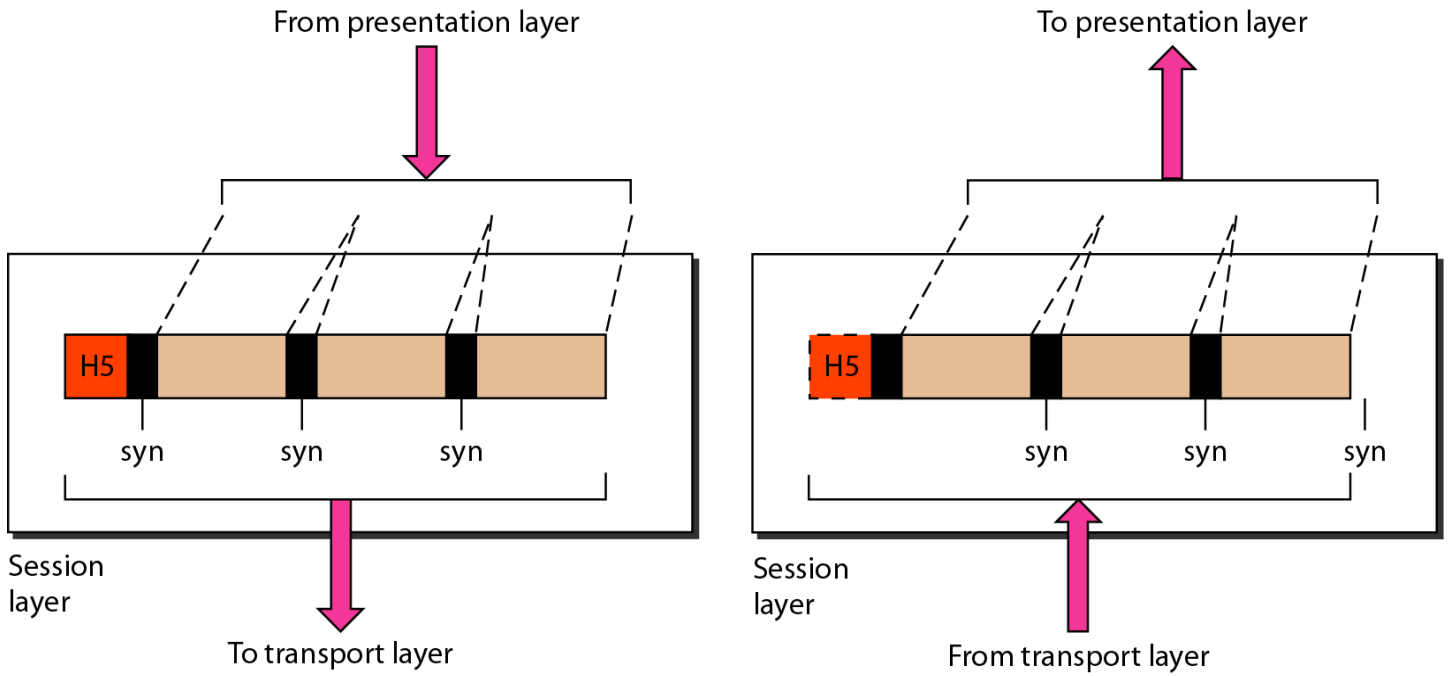


Transport layer



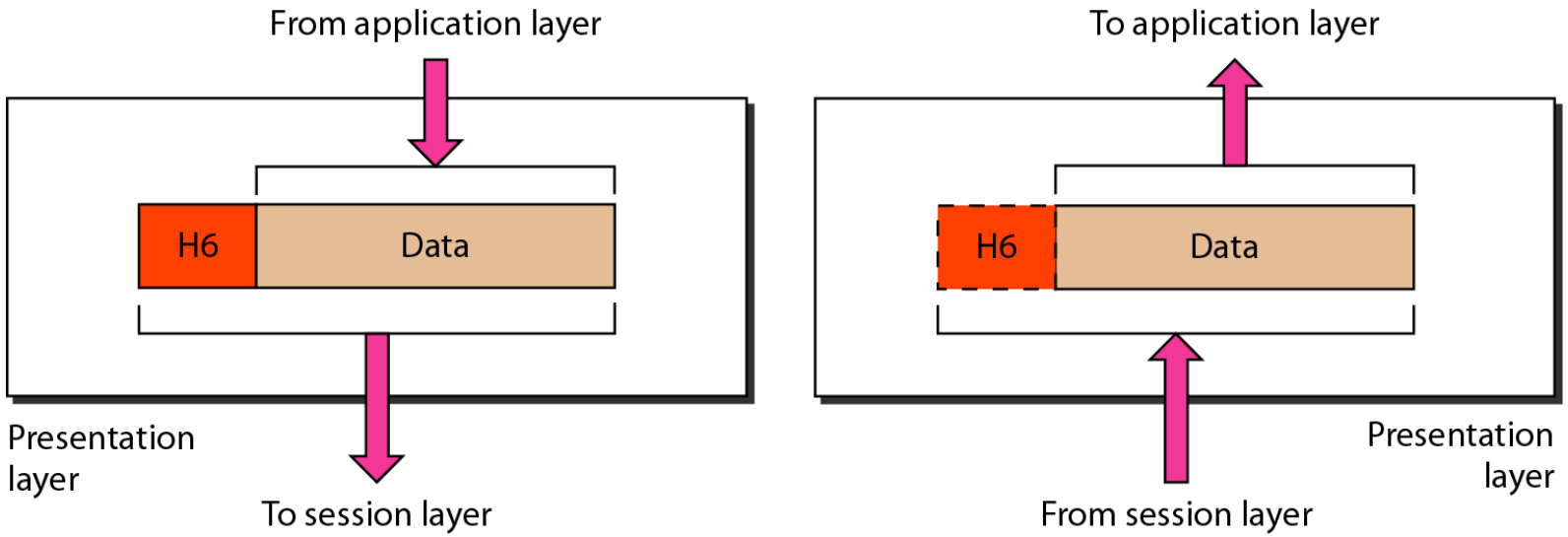
The transport layer is responsible for the delivery of a message from one process to another.

Session layer



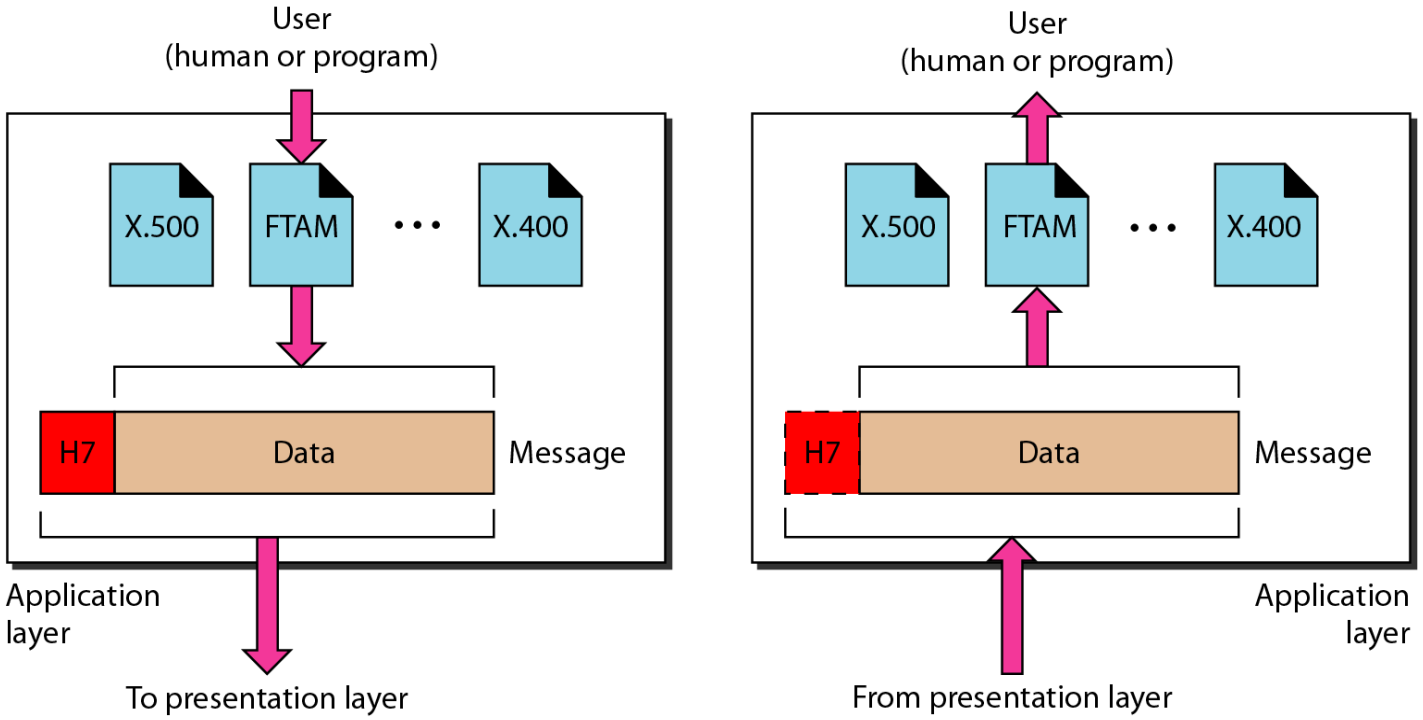
The session layer is responsible for dialog control and synchronization.

Presentation layer



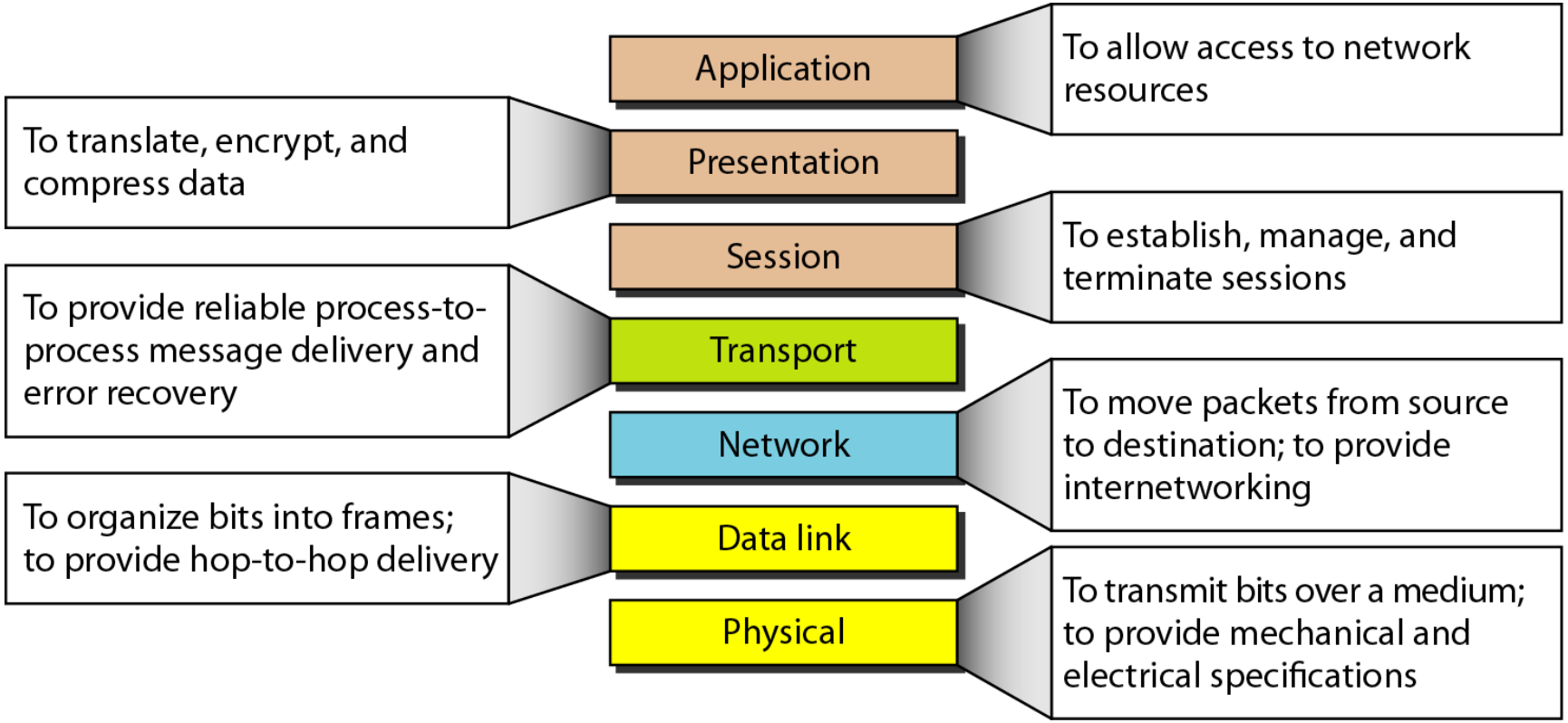
The presentation layer is responsible for translation, compression, and encryption.

Application layer



The application layer is responsible for providing services to the user.

Summary


Application
To allow access to network resources
Presentation
To translate, encrypt, and compress data
Session
To establish, manage, and terminate sessions
Transport
To provide reliable process-to-process message delivery and error recovery
Network
To move packets from source to destination; to provide internetworking
Data link
To organize bits into frames; to provide hop-to-hop delivery
Physical
To transmit bits over a medium; to provide mechanical and electrical specifications

One Minute Paper

Layers of OSI Model

OSI Model

Principles of OSI model

The principles that were applied to arrive at the seven layers can be briefly summarized as follows:

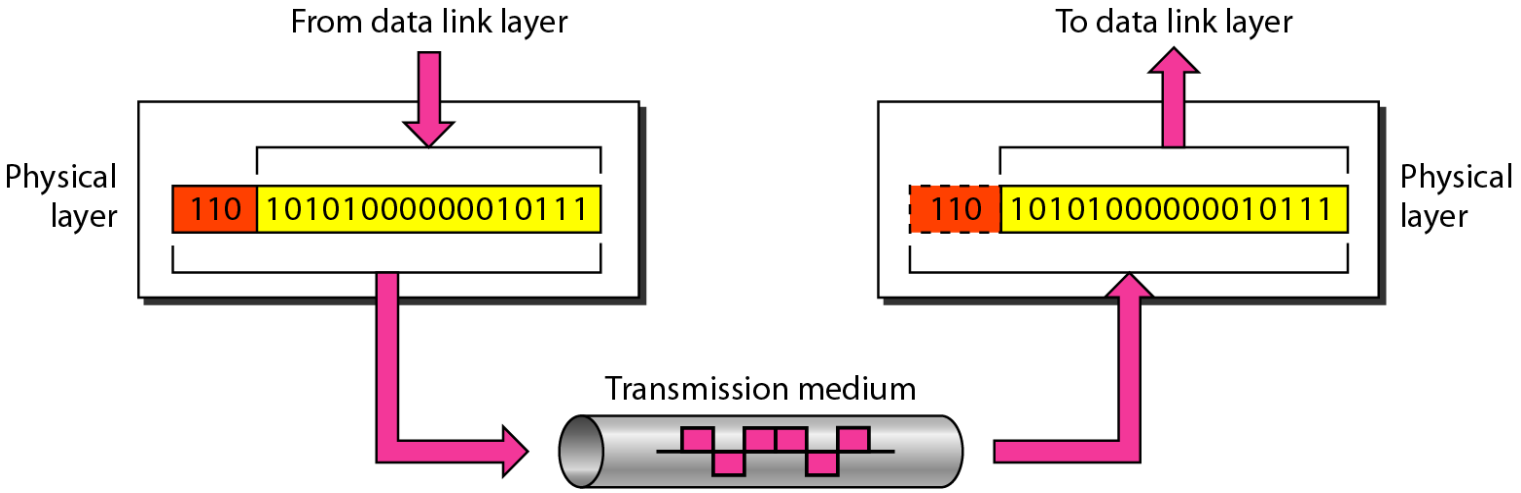
- A layer should be created where a different abstraction is needed.
- Each layer should perform a well-defined function.
- The function of each layer should be chosen with an eye toward defining internationally standardized protocols

Contd…

- The layer boundaries should be chosen to minimize the information flow across the interfaces.
- The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.

Physical Layer

- The physical layer coordinates the functions required to carry a bit stream over a physical medium.
- It deals with the mechanical and electrical specifications of the interface and transmission medium.
- It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to Occur.



Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Contd…

The physical layer is also concerned with the following:

- **Physical characteristics of interfaces and medium.**
  - The physical layer defines the characteristics of the interface between the devices and the transmission medium.
  - It also defines the type of transmission medium.
- **Representation of bits.**
  - data consists of a stream of bits (sequence of 0s or 1s) with no interpretation.
  - To be transmitted, bits must be encoded into signals-electrical or optical.
  - It defines the type of encoding (how 0s and 1s are changed to signals).

Contd…

- **Data rate**
  - The ***transmission rate***-the number of bits sent each second-is also defined by the physical layer.
- **Synchronization of bits**
  - The sender and receiver not only must use the same bit rate.
  - The sender and the receiver clocks must be synchronized.
- **Line configuration**
  - The physical layer is concerned with the connection of devices to the media.
  - In a point-to-point configuration, two devices are connected through a dedicated link.
  - In a multipoint configuration, a link is shared among several devices

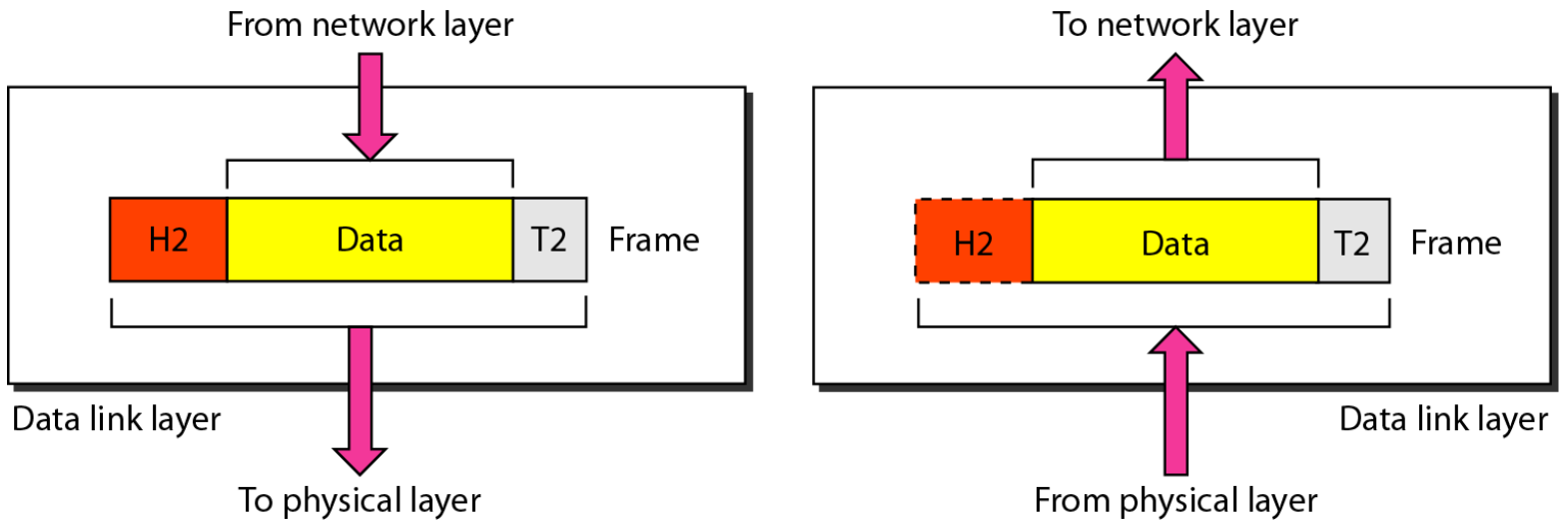
Contd…

- **Physical topology**
  - The physical topology defines how devices are connected to make a network.
  - Devices can be connected by using
    - Mesh topology
    - Star topology
    - Ring topology
    - Bus topology
    - Hybrid topology
- **Transmission mode**
  - The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex.

Data Link Layer

- It makes the physical layer appear error-free to the upper layer (network layer)
- **node – to - node delivery**
  - It is *responsible for transmitting a group of bits* (called frames or packets) to adjacent node without error:

The data link layer is responsible for moving frames from one hop (node) to the next.



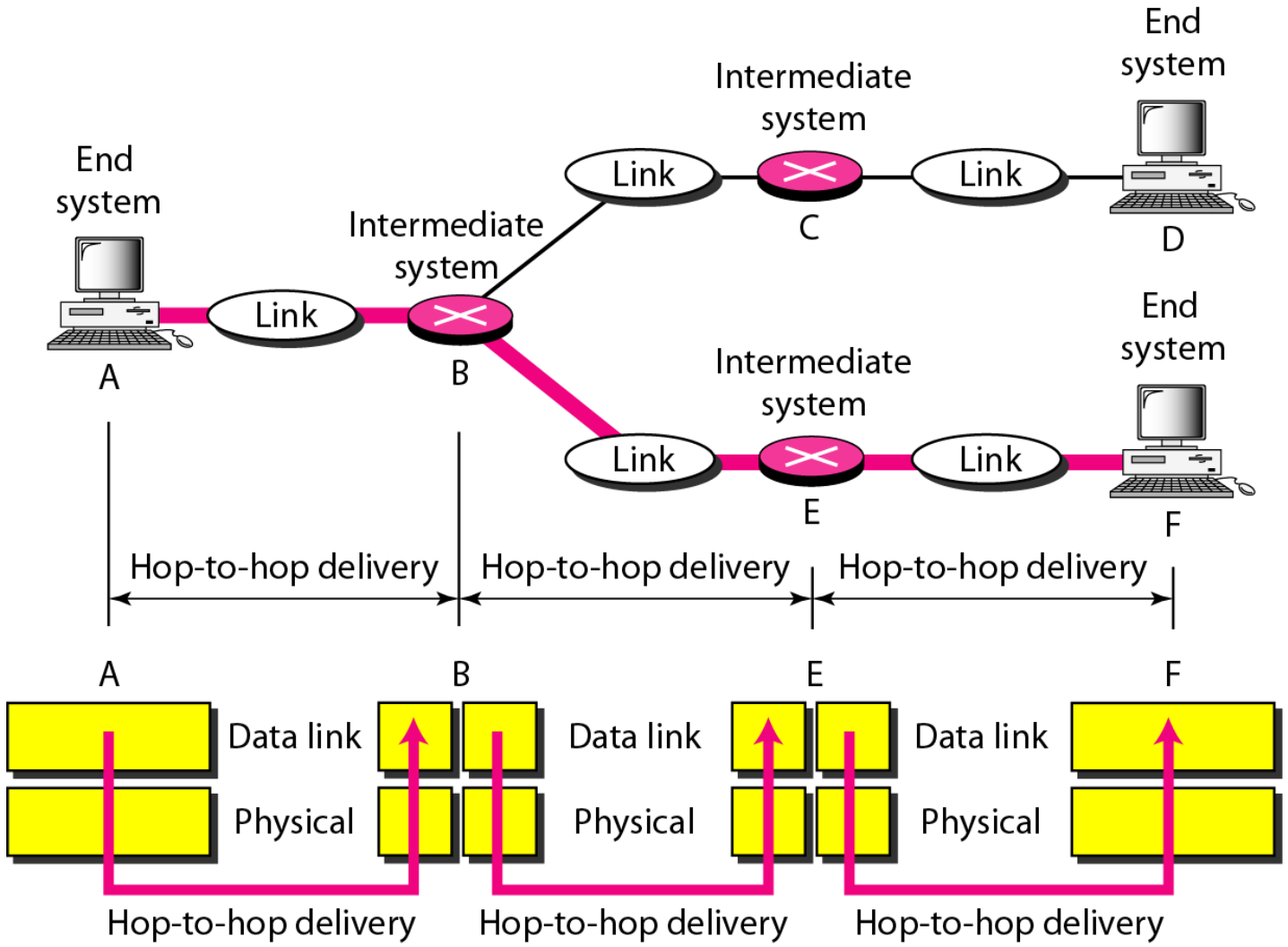
Contd…

- **Framing -** Divides the stream of bits received into data units called frames.
- **Physical addressing** – If frames are to be distributed to different systems on the network, data link layer adds a header to the frame to define the sender and receiver.
- **Flow control** - If the rate at which the data are absorbed by the receiver is less than the rate produced in the sender ,the Data link layer imposes a flow control mechanism.

Contd…

- **Error control** - Used for detecting and retransmitting damaged or lost frames and to prevent duplication of frames. This is achieved through a trailer added at the end of the frame.
- **Access control** -Used to determine which device has control over the link at any given time when 2 or more devices are connected to the same link.

Hop-to-hop delivery



Contd…

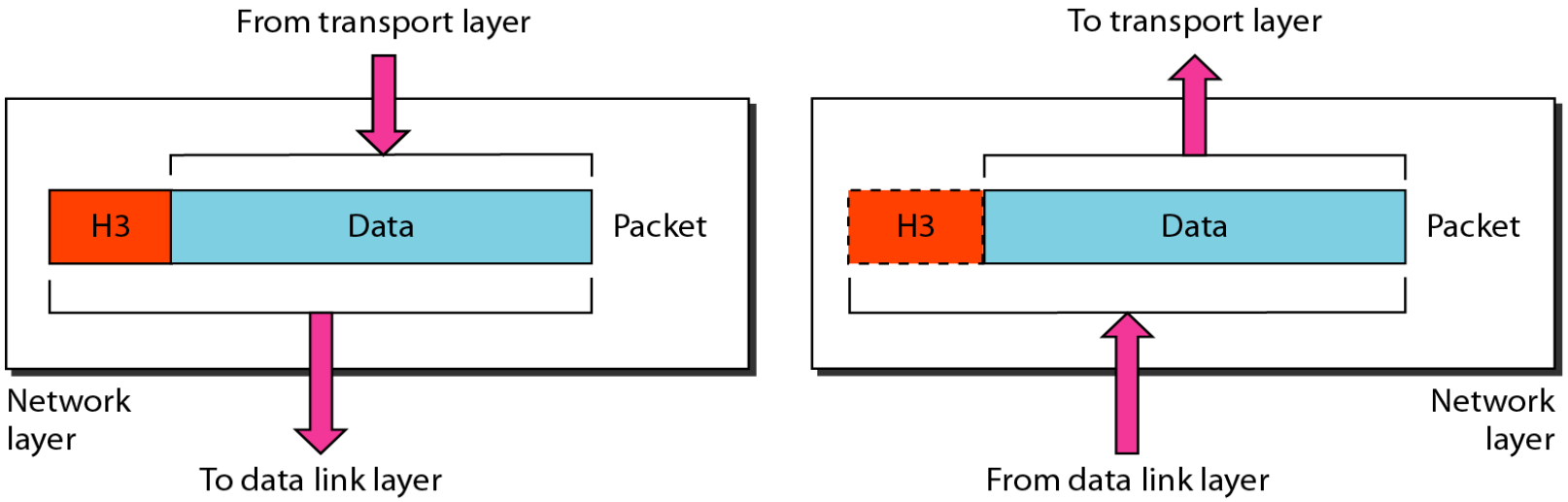
- As the figure shows, communication at the data link layer occurs between two adjacent nodes.
- To send data from A to F, three partial deliveries are made.
- First, the data link layer at A sends a frame to the data link layer at B (a router).
- Second, the data link layer at B sends a new frame to the data link layer at E.
- Finally, the data link layer at E sends a new frame to the data link layer at F.

Contd…

- The frames that are exchanged between the three nodes have different values in the headers.
- The frame from A to B has B as the destination address and A as the source address. Likewise B to E and E to F.
- The values of the trailers can also be different if error checking includes the header of the frame.

Network Layer

- This layer is responsible for the delivery of **packets from source to destination**
- If two systems are connected to the same link, there is usually no need for a network layer.
- However, if the two systems are attached to different networks (links) with connecting devices between the networks (links), there is often a need for the network layer to accomplish source-to-destination delivery.



Contd…

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

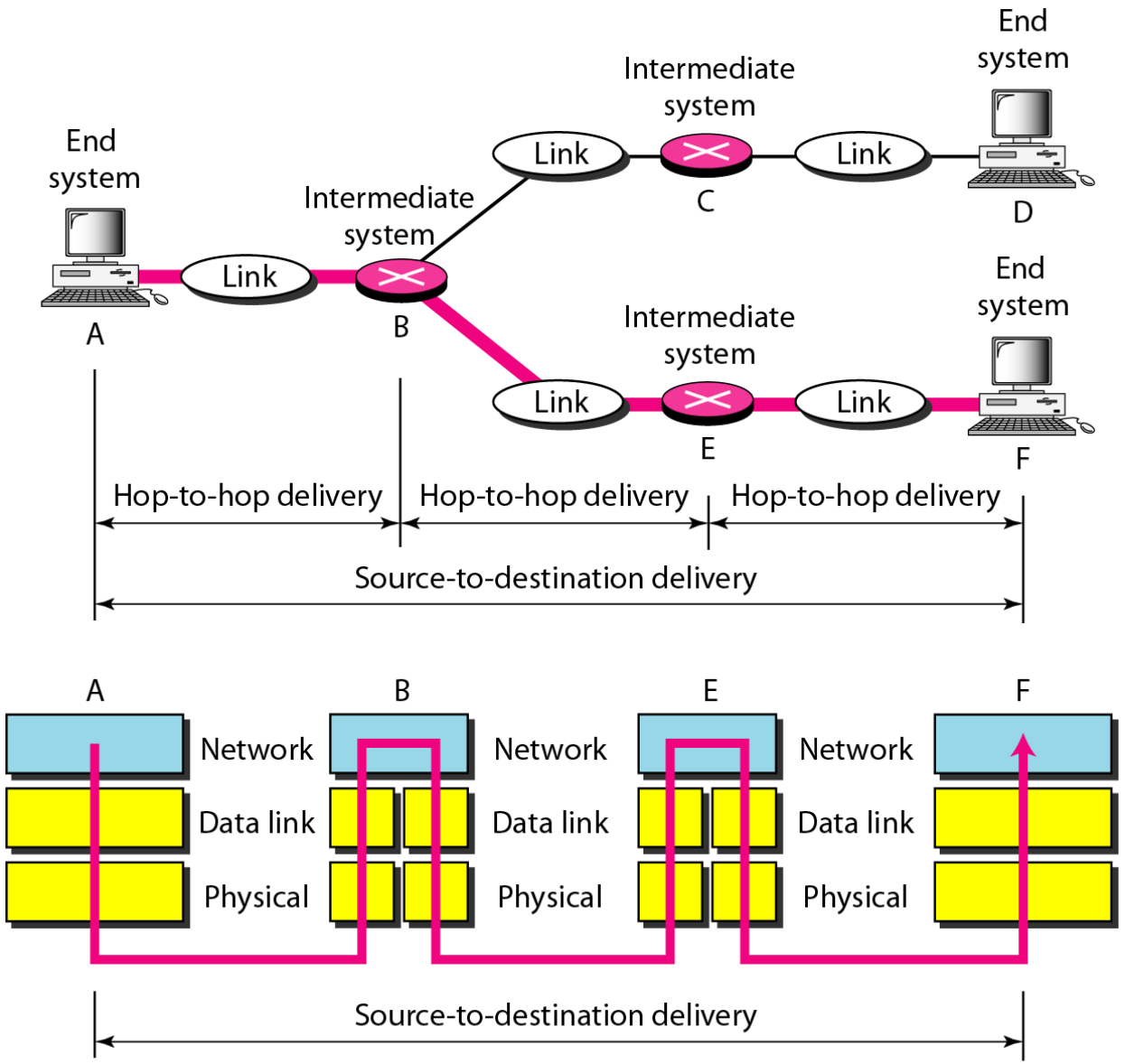
Logical addressing

- If a packet passes the network boundary, we need another addressing system for source and destination called logical address.

Routing

- The devices which connects various independent networks called routers.
- Routers are responsible for delivering packets to final destination.

Source-to-destination delivery



Contd…

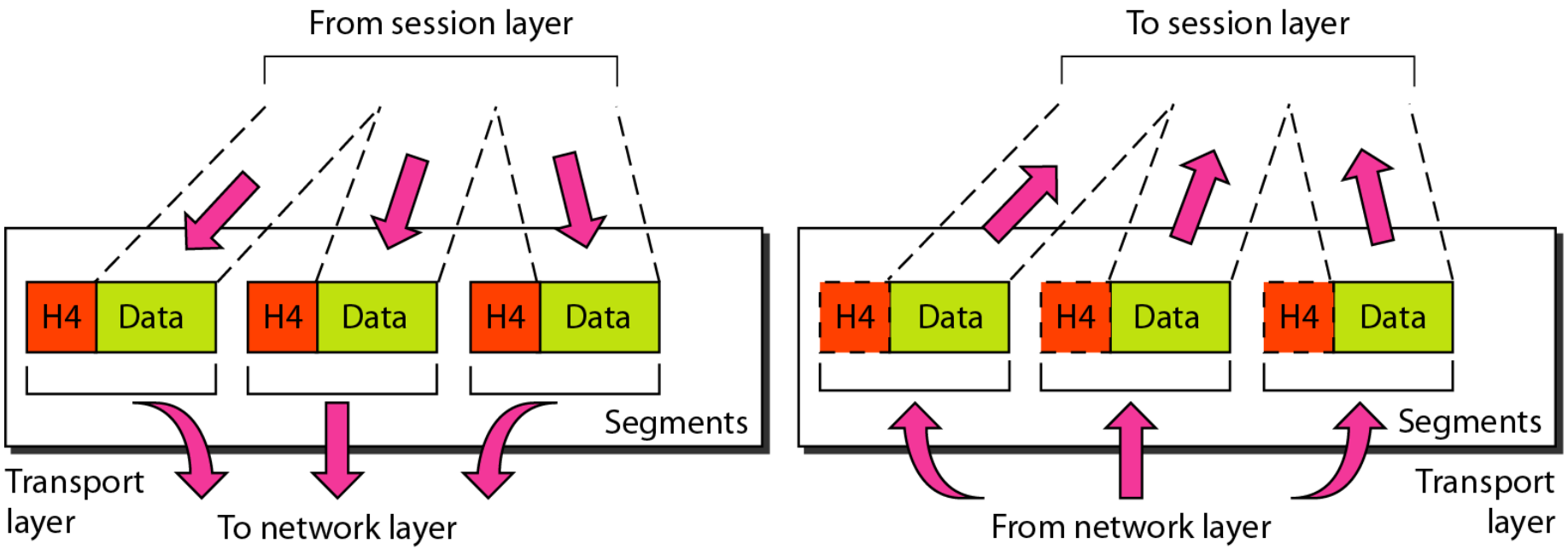
- The network layer at A sends the packet to the network layer at B.
- When the packet arrives at router B, the router makes a decision based on the final destination (F) of the packet.
- router B uses its routing table to find that the next hop is router E.
- The network layer at B, therefore, sends the packet to the network layer at E.
- The network layer at E, in turn, sends the packet to the network layer at F.

Transport Layer

- It is responsible for **process-to-process delivery** of the entire message
- A process is an application program running on a host.

- It does not recognize any relationship between the packets which is arrived from network layer.
- It treats each one independently, as though each piece belonged to a separate message, whether or not it does.
- The transport layer, on the other hand, ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level.

Transport layer



The transport layer is responsible for the delivery of a message from one process to another.

Contd…

- **Port addressing** - The header in this must therefore include a address called port address or service-point address. This layer gets the entire message to the correct process on that computer.

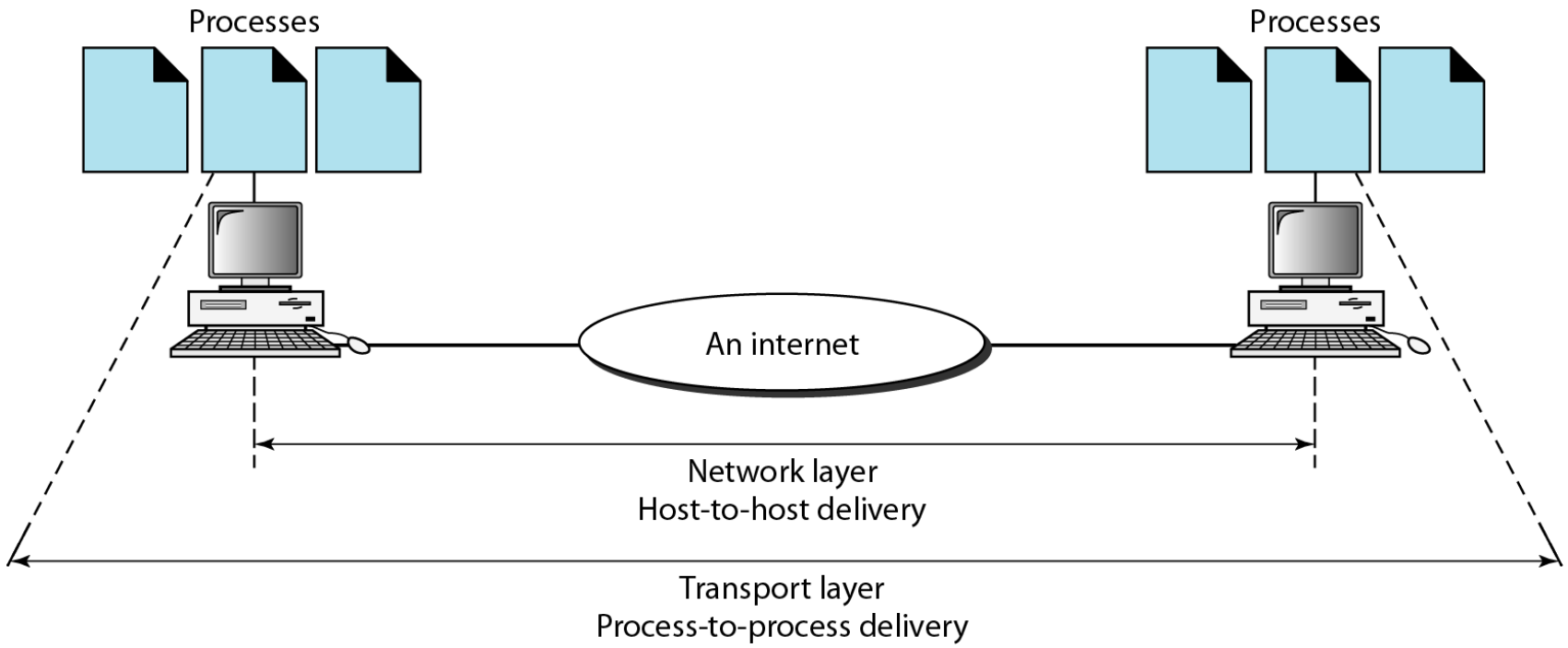
- **Segmentation and reassembly** - The message is divided into segments and each segment is assigned a sequence number. These numbers are arranged correctly on the arrival side by this layer.

Contd…

- **Connection control** - This can either be **connectionless or connection-oriented.** The connectionless treats each segment as a individual packet and delivers to the destination. The connection-oriented makes connection on the destination side before the delivery. After the delivery the connection will be terminated.

- **Flow and error control** - Similar to data link layer, but process to process take place.

Reliable process-to-process delivery of a message

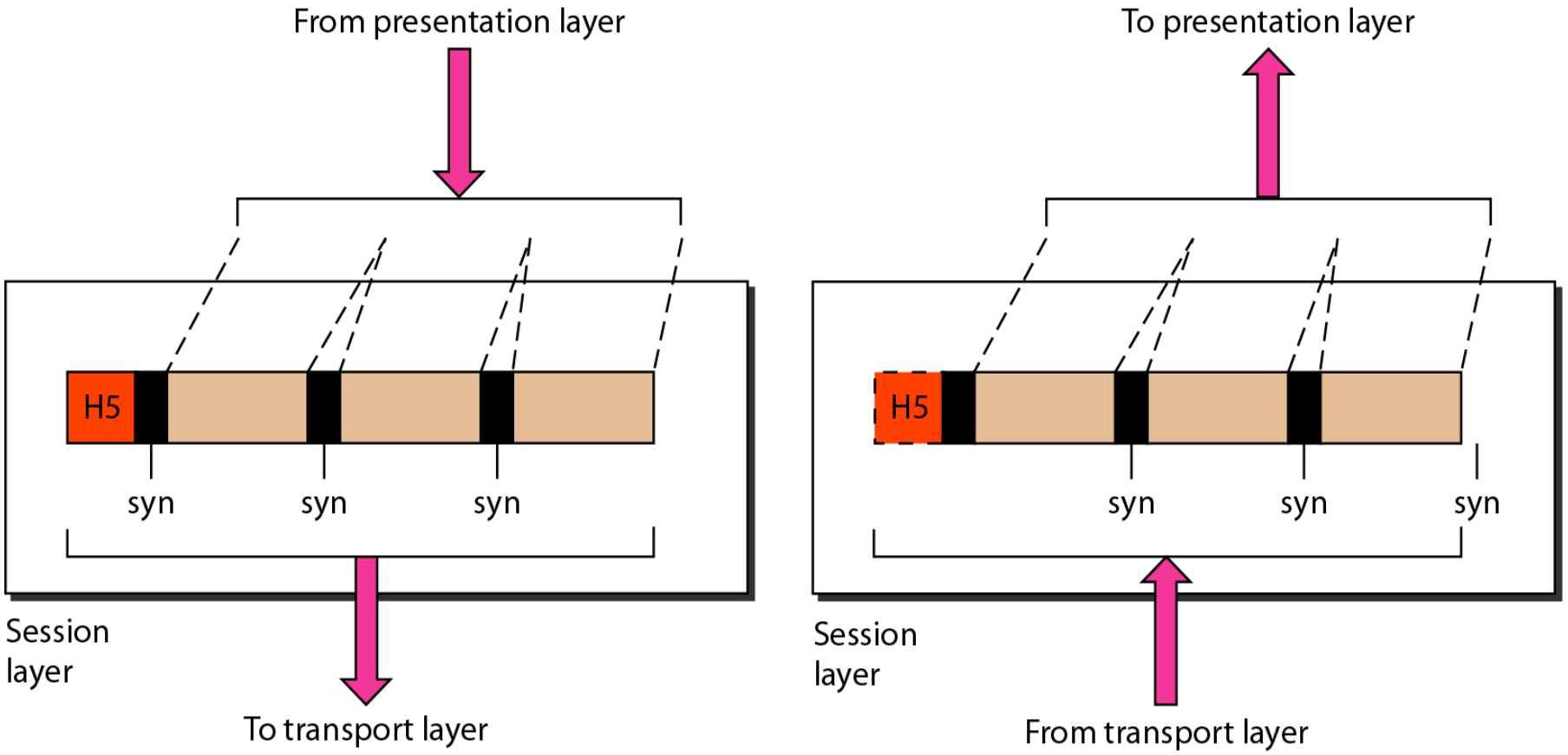


Session Layer

- The services provided by the first three layers (physical, data link, and network) are not sufficient for some processes.
- The session layer is the network *dialog controller*.
- It establishes, maintains, and synchronizes the interaction among communicating systems.
- (i.e) This layer establishes, manages and terminates connections between applications.

The session layer is responsible for dialog control and synchronization.

Session layer



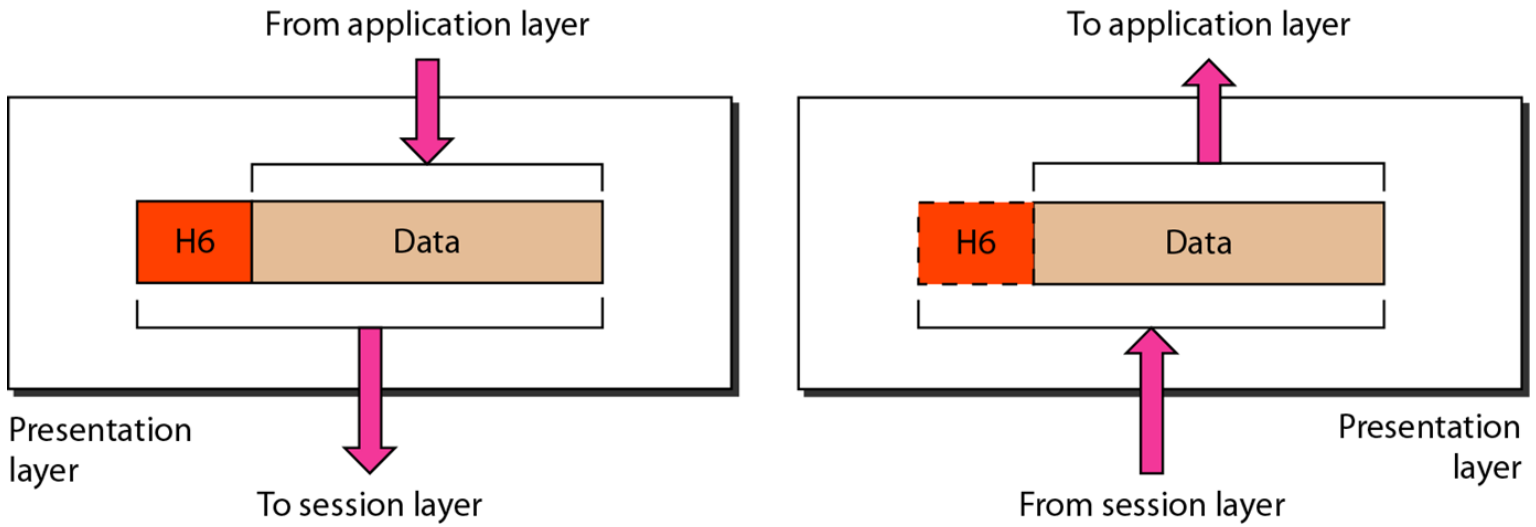
Contd…

- **Dialog control**
  - This session allows two systems to enter into a dialog either in half duplex or full duplex.
- **Synchronization**
  - This allows to add checkpoints (or synchronization points) into a stream of data.
  - For example, if a system is sending a file of 2000 pages, it is advisable to insert checkpoints after every 100 pages to ensure that each 100-page unit is received and acknowledged independently.
  - In this case, if a crash happens during the transmission of page 523, the only pages that need to be resent after system recovery are pages 501 to 523. Pages previous to 501 need not be resent.

Presentation Layer

The presentation layer is responsible for translation, compression, and encryption.

- The presentation layer is concerned with the syntax and semantics of the information exchanged between two systems.



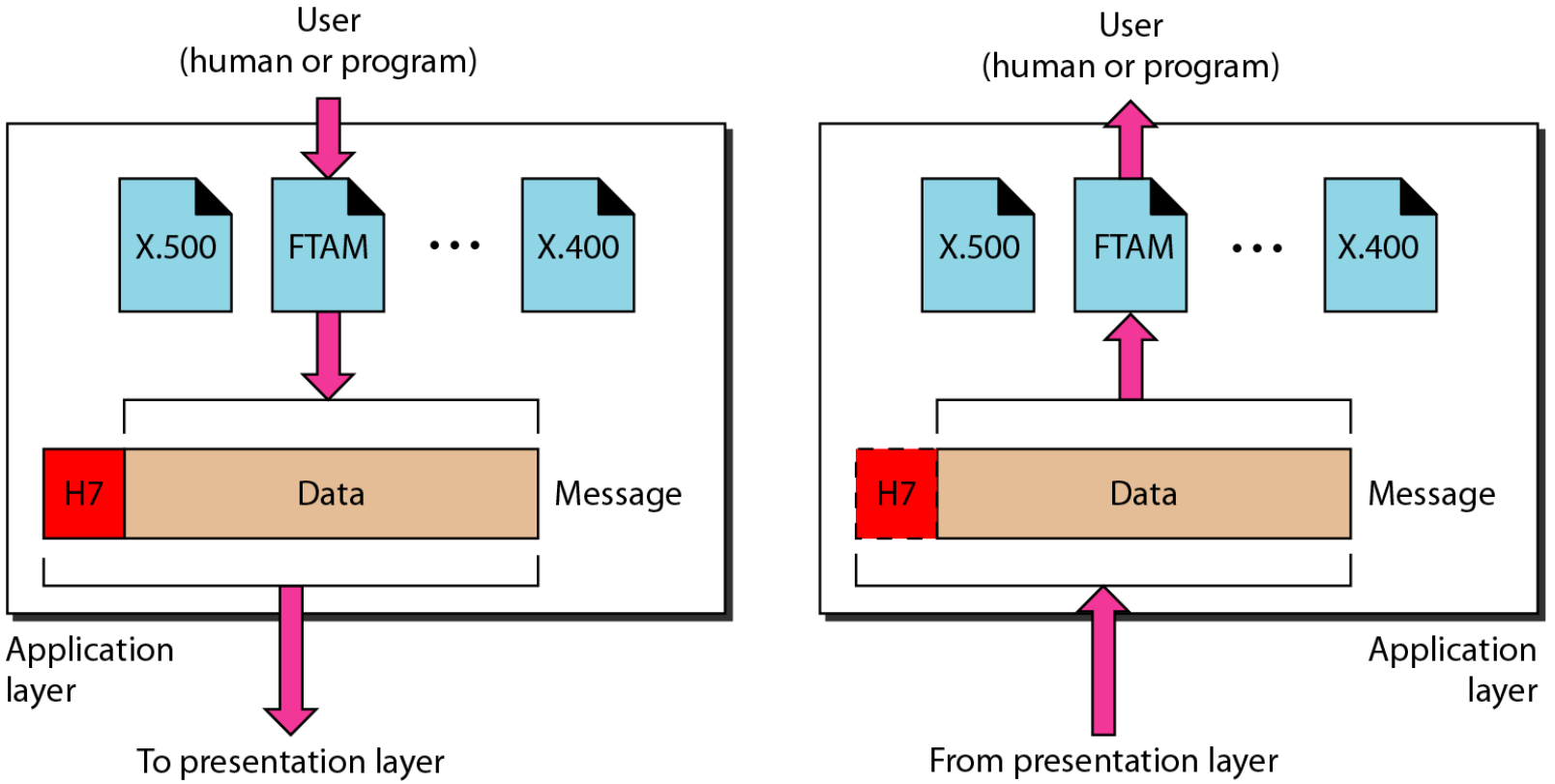
Contd…

- **Translation**
  - Different computers use different encoding system, this layer is responsible for interoperability between these different encoding methods.
  - It will change the message into some common format.
- **Encryption and decryption**
  - It means that sender transforms the original information to another form and sends the resulting message over the network and vice versa.
- **Compression and expansion**
  - Compression reduces the number of bits contained in the information particularly in text, audio and video.

Application Layer

- The application layer enables the user, whether human or software, to access the network.
- It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services.
- Application services
  - X.400 (message handling service)
  - X.500 (Directory services)
  - FTAM (file transfer, access and management)

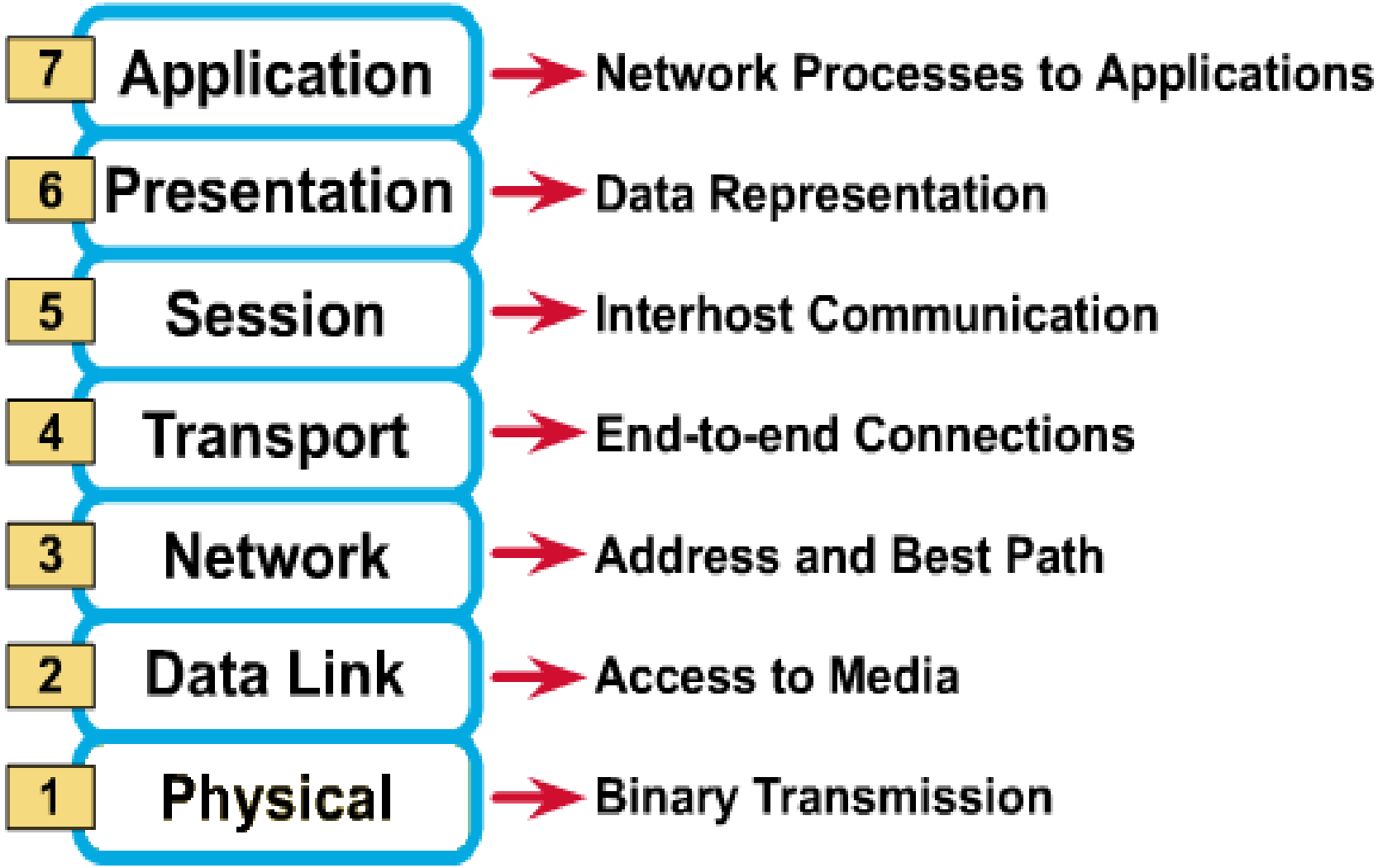
Application layer



The application layer is responsible for providing services to the user.

Contd…

- **FTAM(file transfer, access, mgmt)** - Allows user to access files in a remote host.
- **Mail services** - Provides email forwarding and storage.
- **Directory services** - Provides database sources to access information about various sources and objects.


7
Application
Network Processes to Applications
6
Presentation
Data Representation
5
Session
Interhost Communication
4
Transport
End-to-end Connections
3
Network
Address and Best Path
2
Data Link
Access to Media
1
Physical
Binary Transmission


Thank You
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